

Submersible Motor Pump

Amarex N

50 Hz
DN 50 - DN 100

Type Series Booklet



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Type Series Booklet Amarex N

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Waste Water

Submersible Motor Pump

Amarex N



Main applications

- Waste water management
- Drainage systems
- Sewage treatment plants
- Sludge disposal
- Drainage of rooms and areas at risk of flooding on municipal, commercial and industrial premises

Fluids handled

- Service water
- Grey water
- Waste water with faeces
- Waste water containing long fibres and solid substances
- Fluids containing gas
- Activated sludge
- Digested sludge
- Raw sludge

Operating data

Operating properties

Characteristic	Value
Flow rate	Q Up to 190 m³/h (53 l/s)
Head	H Up to 49 m
Fluid temperature	t Up to +40 °C ¹⁾

Characteristic	Value
Motor rating	P ₂ 0.8 kW to 4.2 kW
Enclosure	IP 68 to EN 60529 / IEC 529

Designation

Example: Amarex N F 80-220 / 04 4 YL G-220

Key to the designation

Code	Description
Amarex N	Type series
F	Impeller type
F	Free-flow impeller
S	Cutter
D	Open, diagonal single-channel impeller
80-220	Size of hydraulic system
04	Motor size
4	Number of motor poles
2	2-pole
4	4-pole
YL	Motor version
UL	Without explosion protection, for fluid temperatures of up to 55 °C
YL ²⁾	With explosion protection, for fluid temperatures of up to 40 °C
WL	Without explosion protection, for fluid temperatures of up to 60 °C
G	Material variant (⇒ Page 10)
G	Pump casing: grey cast iron JL1040 Intermediate casing: grey cast iron JL 1040 Impeller: grey cast iron JL 1040
G1	Pump casing: grey cast iron JL1040 Intermediate casing: grey cast iron JL 1040 Impeller: Noridur 1.4593 (duplex stainless steel)
G2	Pump casing: grey cast iron JL1040 Intermediate casing: grey cast iron JL 1040 Impeller: Norihard 0.9635 (white cast iron)
GH	Pump casing: grey cast iron JL1040 Intermediate casing: Norihard 0.9635 (white cast iron) Impeller: Norihard 0.9635 (white cast iron)
220	Nominal impeller diameter [mm]

Design details

Design

- Fully floodable submersible motor pump
- Not self-priming
- Close-coupled pump set

Shaft seal

- Two bi-directional mechanical seals in tandem arrangement, with liquid reservoir

Impeller types

- Various, application-based impeller types

¹⁾ For short periods (3 to 5 minutes or until the temperature guards trip) UL and WL models can be operated at up to 80 °C.

²⁾ In countries stipulating explosion-proof units for handling sewage with faeces, motor version YL must be used.

D impeller

	Open, diagonal single-vane impeller (impeller type D)	Suitable for the following fluids: fluids containing solid substances and long fibres
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F impeller

	Free-flow impeller (impeller type F)	Suitable for the following fluids: fluids containing solids and stringy material as well as fluids with entrapped air or gas
---	--------------------------------------	--

D and F impellers are suitable for handling the following fluids:

- Activated sludge
- Digested sludge
- Heating sludge
- Mixed water
- Raw waste water
- Raw sludge
- Recirculated sludge

S impeller

	Impeller with cutter (impeller type S)	Suitable for the following fluids: faeces, domestic sewage and waste water containing long fibres
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S impellers are suitable for handling the following fluids:

- Domestic waste water
- Grey water
- Waste water with faeces

Standard bearings

- Grease-lubricated bearings sealed for life
- Maintenance-free

Reinforced bearings (optional, only for S impeller)

The following hydraulic system/motor combinations can be equipped with reinforced bearings:

Reinforced bearings

Size of hydraulic system	Motor size and number of poles
Amarex N S 50-172 (motor version YL)	002, 012, 022
Amarex N S 50-222 (motor version YL)	032, 042

Pump-end bearings:

- Grease-lubricated bearing sealed for life

Drive

- Three-phase asynchronous squirrel-cage motor

Explosion-proof pump sets comply with Ex dc IIB type of protection.

Materials

Overview of available materials

Component	Material
Casing	JL 1040
Intermediate casing	JL 1040
Impeller	JL 1040 ³⁾
Cutter (only Amarex N S)	1.2080 (K100)
Shaft	1.4021
Mechanical seal (motor end)	Carbon/Al ₂ O ₃
Mechanical seal (pump end)	SiC/SiC
Screws/bolts	A2
Sealing elements	NBR

Product benefits

- Easy and fast installation/removal with polarised, absolutely watertight cable and KSB plug connection ensuring correct installation
- Motor absolutely watertight, also in the event of damage to the cable sheath, with individually stripped, tinned and resin-sealed conductors
- Highest operating reliability with optimum motor selection for operating mode S1, thermal class F, explosion-proof variant in Ex d IIB T4 Gb
- Long service life with shaft made of corrosion-resistant stainless steel
- Zero maintenance, ideal for continuous duty, with long-life bearings sealed on both sides, grease-packed for life
- Environmentally friendly, non-toxic, food-approved oil
- Significant reduction of energy costs by optimised hydraulic system and high efficiency
- Absolutely reliable and non-clogging design with optimised cutter (S impeller).
- No risk of leakage with close-coupled pump design
- Stationary pump sets easy to install and remove with automatic, bolt-free connection; leakage prevented by elastic sealing elements
- Ease of service with wetted bolts made of stainless steel, which are easy to undo even after years of operation
- Two bi-directional mechanical seals with oil reservoir filled with ecologically acceptable oil provide double safety
- Well suited for mechanical seal with covered spring for abrasive and aggressive fluids
- Only one set of spare parts for all pump sizes thanks to modular system

Certifications

Label	Effective in:	Note
	Europe	Suitable for plants to Standard EN 12050-1
	Europe	Amarex N DN 50 - DN 100 BMW (test report) Nos.: 0420266-01 to 05

³⁾ Amarex N F 50/65/80/100 in material variant G1 = 1.4593 (Noridur), G2 = 0.9635 (Norihard)

Product overview / selection tables

detailed advice is available from our specialist department.
 Make use of our laboratory's wealth of experience when selecting materials.

Table of fluids handled

The table below for your guidance is based on KSB's long-standing experience. The data are standard values and are not to be considered as generally binding recommendations. More

Selection aid for materials and hydraulic systems per fluid

Fluid handled ⁴⁾	Recommended material	Recommended impeller type ⁵⁾	Comments, further recommendations
Grey water	JL 1040	F, D, S	Free passage > any solids contained, possibly pre-screened
River water	JL 1040	F, D	Free passage > any solids contained, possibly pre-screened
Contaminated surface water	JL 1040	F, D	Free passage > any solids contained, possibly pre-screened
Waste water			
▪ Untreated municipal waste water	JL 1040	F, D, S	ATV ⁶⁾ recommends a free passage of 100 mm; min. free passage: 76 mm
▪ Waste water containing air or gas	JL 1040	F	Up to 8 %, contact KSB for handling fluids with high outgassing rates
▪ Raw waste water	JL 1040	F, D	ATV ⁶⁾ recommends a free passage of 100 mm; min. free passage: 76 mm
▪ Mixed water	JL 1040	F	Free passage > any solids contained, possibly pre-screened
▪ Waste water or grey water containing long fibres	JL 1040	F, D, S	Free passage > any solids contained, possibly pre-screened
▪ Highly abrasive waste water causing wear (chemically neutral)	Norihard	F	For solids contents < 5 g/l material variants G2, GH
▪ Corrosive waste water	Noridur	F	Material variant G1 if required according to analysis of the fluid handled
Sludge			
▪ Raw sludge	JL 1040	D, F	Pumpable up to a dry substance content of: 13 % (D), 8 % (F)
▪ Digested sludge	JL 1040	D, F	Pumpable up to a dry substance content of: 13 % (D), 8 % (F)
▪ Activated sludge	JL 1040	D, F	Pumpable up to a dry substance content of: 13 % (D), 8 % (F)
Industrial waste water containing ...			
▪ Paint suspensions	JL 1040	F	Solvent-free, observe the operator's instructions!
▪ Lacquer/paint/varnish suspensions	JL 1040	F	Solvent-free, contact KSB for silicone-free version
▪ Fibres/pulp	JL 1040	F, D, S	
▪ Chips/swarf	Norihard	F	Material variant G2 or GH, special mechanical seal; solids content < 5 g/l
▪ Abrasive substances ⁷⁾	Norihard	F	Material variant G2 or GH, special mechanical seal; solids content < 5 g/l
Mildly acidic industrial waste water	JL 1040	F	pH value ≥ 6.5: G1 variant and FPM (Viton) O-rings
Non-corrosive waste water			
▪ Ammonium hydroxide	JL 1040	F	
▪ Ammonium hydroxide 5 % NH ₄ OH	JL 1040	F	
▪ Urea 25 % (NH ₂) ₂ -CO	JL 1040	F	
▪ Potassium hydroxide 10 % KOH	JL 1040	F	
▪ Calcium hydroxide 5 % Ca(OH) ₂	JL 1040	F	
▪ Sodium hydroxide 5 % NaOH	JL 1040	F	
▪ Sodium carbonate 30 % Na ₂ CO ₃	JL 1040	F	

4) For any fluids which are not listed in this table contact KSB.

5) The first impeller type listed should be given preference.

6) ATV = German regulatory body for waste water management

7) Severe hydroabrasive wear occurs if solids contents of approx. 0.5 g/l or higher are combined with circumferential speeds exceeding 20 m/s or low-flow conditions to the left of the duty point.

Fluid handled ⁴⁾	Recommen- ded material	Recomm- ended impeller type ⁵⁾	Comments, further recommendations
Non-corrosive waste water containing ...			
▪ Aliphatic hydrocarbons, e.g. oils, petrol, butane, methane	JL 1040	F	FPM (Viton) O-rings, TEHSITE cable, for high concentrations contact KSB.
▪ Aromatic hydrocarbons, e.g. benzene, styrene	JL 1040	F	FPM (Viton) O-rings, TEHSITE cable, for high concentrations contact KSB.
▪ Chlorinated hydrocarbons (e.g. tetrachloroethylene, ethylene chloride, chloroform, methylene chloride)	JL 1040	F	FPM (Viton) O-rings, TEHSITE cable, for high concentrations contact KSB.
Highly abrasive industrial waste water causing wear (chemically neutral)⁸⁾			
▪ Lime water	Norihard	F	For sinter contents < 5 g/l, material variant GH
▪ Lime milk containing quartz and pigment suspension	Norihard	F	Lime milk of up to 15 %, material variant GH
▪ Water/sand mixture	Norihard	F	Up to 5 g/l solids content, material variant GH

4) For any fluids which are not listed in this table contact KSB.

5) The first impeller type listed should be given preference.

8) The material variants required highly depend on the operating hours, rotational speed and flow velocity.

Overview of product features

Standard designs

Size	S impeller	D impeller	F impeller
	Amarex N S 50-...	Amarex N D 80-... Amarex N D 100-...	Amarex N F 50-... Amarex N F 65-... Amarex N F 80-... Amarex N F 100-...
Material variant	G	G	G
Number of motor poles			
2-pole	50-172/... 50-222/...	-	50-170/... 50-220/... 65-170/...
4-pole	-	80-220/... 100-220/...	65-220/... 80-220/... 100-200/...
Explosion protection			
Motor version UL	Non-explosionproof		
Motor version YL ⁹⁾	Ⓔ Ex d IIB T4 Gb		
Motor version WL	Non-explosionproof		
Motor			
Starting method	DOL ¹⁰⁾		
Voltage	400 V		
Cooling	Cooled by surrounding fluid		
Mode of operation	S1: submerged (max. 25 m) (see ²⁾ in outline drawing) S3: outside the fluid (see ¹⁾ in outline drawing)		
Power cable			
Type	Rubber-sheathed cable (H07RN8-F 7G1.5 ²⁾)		
Length	10 m		
Cable entry	Absolutely watertight		
Sealing elements			
Shaft seal	Mechanical seal		
Elastomer seals	NBR		
Monitoring equipment			
Winding temperature version YL	Temperature monitoring circuit (with automatic reset and start-up): bimetal switch directly connected with the control circuit of the motor contactor Limiting circuit (temperature limit for explosion protection without automatic reset): bimetal switch connected via a tripping unit with manual reset		
Winding temperature version UL, WL	Temperature monitoring circuit (with automatic reset and start-up): bimetal switch directly connected with the control circuit of the motor		
Coating	Environmentally-friendly KSB top coat (two-component epoxy paint), colour RAL 5002, film thickness = 80 µm		
Installation (⇒ Page 37)			
Stationary, with guide hoop	Installation depths 1.5 m/1.8 m/2.1 m		
Stationary, with single guide rail	Installation depth 4.5 m		
Stationary, with twin guide rail	Installation depth 4.5 m		
Stationary, with guide wire	Installation depth 4.5 m		
Transportable	Installation depth 4.5 m		
Maximum temperature of fluid handled			
Motor version UL	55 °C		
Motor version YL	40 °C		
Motor version WL	60 °C		

⁹⁾ In countries stipulating explosion-proof units for handling sewage with faeces, motor version YL must be used.

¹⁰⁾ Maximum switching frequency: 30 starts per hour

Standard variants

Size	S impeller	D impeller	F impeller		
	Amarex N S 50-172/... Amarex N S 50-222/...	Amarex N D 80-220/... Amarex N D 100-220/...	Amarex N F 50-... Amarex N F 65-... Amarex N F 80-... Amarex N F 100-...		
Material variants	G	G	G1	G2	GH
Number of motor poles					
2-pole	-	-	50-170/... 50-220/... 65-170/...		
4-pole	-	-	65-220/... 80-220/... 100-200/...		
Shaft material					
Material 1.4462 + C45 N	X	X	X		
Motor					
Voltage	230 V, 415 V, 500 V, 690 V				
Monitoring equipment					
Leakage sensor in the motor space ¹¹⁾	X	X	X		
Suction flange					
Drilled to DIN/ISO PN16 or ASME 150 lb	-	-	X		
Sealing elements					
Elastomers: O-rings and flange gaskets made of Viton, lower mechanical seal with Viton gaskets	X	X	X		
Shaft seal: special mechanical seal (mechanical seal with covered spring HJ977) ¹²⁾	X	X	X		
Power cables					
Standard rubber-sheathed cable (H07RN8-F 7G1.5 ²⁾ ; for versions ULG, YLG, WLG ¹³⁾	X	X	X		
Standard rubber-sheathed cable (H07RN8-F 8x1.5 ²⁾ for pump with leakage sensor ¹³⁾	X	X	X		
TEHSITE power cable (8G1.5) for pump with or without leakage sensor, for versions ULG, YLG, WLG ¹⁴⁾	X	X	X		
Shielded rubber-sheathed cable (S07RC4N8-F-8G1.5) for pump with or without leakage sensor, for versions ULG, YLG, WLG operated with a frequency inverter ¹⁴⁾	X	X	X		
Coating	Environmentally-friendly KSB standard coating (two-component epoxy paint), colour RAL 5002, film thickness = 300 µm				
Installation (⇒ Page 37)					
Stationary, with guide hoop	Installation depths 1.5 m/1.8 m/2.1 m				
Stationary, with single guide rail	Installation depth 6.0 m				
Stationary, with twin guide rail	Installation depth 6.0 m				
Stationary, with guide wire	Installation depth 9.5 m				
Transportable	Installation depth 4.5 m				

¹¹⁾ 8-core connection cable required

¹²⁾ Silicon carbide/silicon carbide seal faces, Viton sealing elements, stainless steel spring and metal part, Viton (FPM) O-rings and flange gaskets

¹³⁾ Total lengths available 15 m/20 m/30 m/40 m/50 m

¹⁴⁾ Total lengths available 10 m/15 m/20 m/30 m/40 m/50 m

Material variants

Overview of material variants for standard models

Component		S impeller	D impeller	F impeller
		Amarex N S 50-172/... Amarex N S 50-222/...	Amarex N D 80-220/... Amarex N D 100-220/...	Amarex N F 50-... Amarex N F 65-... Amarex N F 80-... Amarex N F 100-...
		G	G	G
Casing		JL 1040	JL 1040	JL 1040
Intermediate casing		JL 1040	JL 1040	JL 1040
Impeller		JL 1040	JL 1040	JL 1040
Cutter		1.2080.02 (K100)	-	-
Shaft		1.4021	1.4021	1.4021
Mechanical seal	Drive end	Carbon/Al ₂ O ₃	Carbon/Al ₂ O ₃	Carbon/Al ₂ O ₃
	Pump end	SiC/SiC	SiC/SiC	SiC/SiC
Screws/bolts		A2	A2	A2
Elastomer seals		NBR	NBR	NBR

Overview of material variants for standard models

Component		S impeller	D impeller	F impeller		
		Amarex N S 50-172/... Amarex N S 50-222/...	Amarex N D 80-220/... Amarex N D 100-220/...	Amarex N F 50-... Amarex N F 65-... Amarex N F 80-... Amarex N F 100-...		
		G	G	G1	G2	GH
Casing		-	-	JL 1040	JL 1040	JL 1040
Intermediate casing		-	-	JL 1040	JL 1040	0.9635 ¹⁶⁾
Impeller		-	-	1.4593 ¹⁵⁾	0.9635 ¹⁶⁾	0.9635 ¹⁶⁾
Cutter		-	-	-	-	-
Shaft		-	-	1.4021	1.4021	1.4021
Mechanical seal	Drive end	-	-	Carbon/Al ₂ O ₃	Carbon/Al ₂ O ₃	Carbon/Al ₂ O ₃
	Pump end	-	-	SiC/SiC	SiC/SiC	SiC/SiC
Screws/bolts		-	-	A2	A2	A2
Elastomer seals		-	-	NBR ¹⁷⁾	NBR ¹⁷⁾	NBR ¹⁷⁾

¹⁵⁾ Noridur (= duplex stainless steel)

¹⁶⁾ Norihard (= white cast iron)

¹⁷⁾ Optional: FPM

Technical data

Impeller diameter	Size	Motor version	P ₁	P ₂	I _N	I _A	Fluid temperature	[kg]	Mat. No.
			[kW]	[kW]	[A]	[A]	[°C]		
120	S 50-172/002 LG-120	ULG	1,75	1,30	3,56	20	55	47	39100017
	S 50-172/002 LG-120	YLG	1,75	1,30	3,56	20	40	47	39100018
	S 50-172/002 LG-120	WLG	1,75	1,30	3,56	20	60	47	
140	S 50-172/002 LG-140	ULG	1,75	1,30	3,56	20	55	47	39100019
	S 50-172/002 LG-140	YLG	1,75	1,30	3,56	20	40	47	39100020
	S 50-172/002 LG-140	WLG	1,75	1,30	3,56	20	60	47	
160	S 50-172/012 LG-160	ULG	2,60	1,90	4,50	20	55	47	39100021
	S 50-172/012 LG-160	YLG	2,60	1,90	4,50	20	40	47	39100022
	S 50-172/002 LG-160	WLG	2,60	1,90	4,50	20	60	47	
175	S 50-222/032 LG-175	ULG	4,00	3,10	7,00	50	55	58	39100041
	S 50-222/032 LG-175	YLG	4,00	3,10	7,00	50	40	58	39100042
	S 50-222/032 LG-175	WLG	4,00	3,10	7,00	50	55	58	
190	S 50-222/042 LG-190	ULG	5,30	4,20	8,80	50	55	58	39100043
	S 50-222/042 LG-190	YLG	5,30	4,20	8,80	50	40	58	39100044
	S 50-222/042 LG-190	WLG	5,30	4,20	8,80	50	55	58	
90	F 50-170/002 LG-90	ULG	1,75	1,30	3,56	20	55	41	39100045
	F 50-170/002 LG-90	YLG	1,75	1,30	3,56	20	40	41	39100046
	F 50-170/002 LG-90	WLG	1,75	1,30	3,56	20	60	41	
107	F 50-170/002 LG-107	ULG	1,75	1,30	3,56	20	55	41	39100047
	F 50-170/002 LG-107	YLG	1,75	1,30	3,56	20	40	41	39100048
	F 50-170/002 LG-107	WLG	1,75	1,30	3,56	20	60	41	
120	F 50-170/012 LG-120	ULG	2,60	1,90	4,50	20	55	42	39100049
	F 50-170/012 LG-120	YLG	2,60	1,90	4,50	20	40	42	39100050
	F 50-170/012 LG-120	WLG	2,60	1,90	4,50	20	60	42	
130	F 50-170/022 LG-130	ULG	3,06	2,30	5,10	20	55	42	39100051
	F 50-170/022 LG-130	YLG	3,06	2,30	5,10	20	40	42	39100052
	F 50-170/022 LG-130	WLG	3,06	2,30	5,10	20	60	43	
140	F 50-170/022 LG-140	ULG	3,06	2,30	5,10	20	55	43	39100053
	F 50-170/022 LG-140	YLG	3,06	2,30	5,10	20	40	43	39100054
	F 50-170/022 LG-140	WLG	3,06	2,30	5,10	20	60	43	
130	F 50-220/032 LG-130	ULG	4,00	3,10	7,00	50	55	52	39100067
	F 50-220/032 LG-130	YLG	4,00	3,10	7,00	50	40	52	39100068
	F 50-220/032 LG-130	WLG	4,00	3,10	7,00	50	60	52	
140	F 50-220/032 LG-140	ULG	4,00	3,10	7,00	50	55	52	39100069
	F 50-220/032 LG-140	YLG	4,00	3,10	7,00	50	40	52	39100070
	F 50-220/032 LG-140	WLG	4,00	3,10	7,00	50	60	52	
150	F 50-220/042 LG-150	ULG	5,30	4,20	8,80	50	55	53	39100071
	F 50-220/042 LG-150	YLG	5,30	4,20	8,80	50	40	53	39100072
	F 50-220/042 LG-150	WLG	5,30	4,20	8,80	50	60	53	
160	F 50-220/042 LG-160	ULG	5,30	4,20	8,80	50	55	53	39100073
	F 50-220/042 LG-160	YLG	5,30	4,20	8,80	50	40	53	39100074
	F 50-220/042 LG-160	WLG	5,30	4,20	8,80	50	60	53	
170	F 50-220/042 LG-170	ULG	5,30	4,20	8,80	50	55	54	39100075
	F 50-220/042 LG-170	YLG	5,30	4,20	8,80	50	40	54	39100076
	F 50-220/042 LG-170	WLG	5,30	4,20	8,80	50	60	53	
180	F 50-220/042 LG-180	ULG	5,30	4,20	8,80	50	55	54	39100077
	F 50-220/042 LG-180	YLG	5,30	4,20	8,80	50	40	54	39100078
	F 50-220/042 LG-180	WLG	5,30	4,20	8,80	50	60	53	
120	F 65-170/032 LG-120	ULG	4,00	3,10	7,00	50	55	58	39100085
	F 65-170/032 LG-120	YLG	4,00	3,10	7,00	50	40	58	39100086
	F 65-170/032 LG-120	WLG	4,00	3,10	7,00	50	60	58	
128	F 65-170/032 LG-128	ULG	4,00	3,10	7,00	50	55	58	39100087
	F 65-170/032 LG-128	YLG	4,00	3,10	7,00	50	40	58	39100088
	F 65-170/032 LG-128	WLG	4,00	3,10	7,00	50	60	58	
136	F 65-170/032 LG-136	ULG	4,00	3,10	7,00	50	55	59	39100089
	F 65-170/032 LG-136	YLG	4,00	3,10	7,00	50	40	59	39100090
	F 65-170/032 LG-136	WLG	4,00	3,10	7,00	50	60	58	
146	F 65-170/042 LG-146	ULG	5,30	4,20	8,80	50	55	59	39100091

Impeller diameter	Size	Motor version	P ₁	P ₂	I _N	I _A	Fluid temperature	[kg]	Mat. No.
			[kW]	[kW]	[A]	[A]	[°C]		
	F 65-170/042 LG-146	YLG	5,30	4,20	8,80	50	40	59	39100092
	F 65-170/042 LG-146	WLG	5,30	4,20	8,80	50	60	59	
152	F 65-170/042 LG-152	ULG	5,30	4,20	8,80	50	55	60	39100093
	F 65-170/042 LG-152	YLG	5,30	4,20	8,80	50	40	60	39100094
	F 65-170/042 LG-152	WLG	5,30	4,20	8,80	50	60	59	
158	F 65-170/042 LG-158	ULG	5,30	4,20	8,80	50	55	60	39100095
	F 65-170/042 LG-158	YLG	5,30	4,20	8,80	50	40	60	39100096
	F 65-170/042 LG-158	WLG	5,30	4,20	8,80	50	60	59	
112	F 65-220/004 LG-112	ULG	1,23	0,80	2,75	17,4	55	49	39100097
	F 65-220/004 LG-112	YLG	1,23	0,80	2,75	17,4	40	49	39100098
	F 65-220/004 LG-112	WLG	1,23	0,80	2,75	17,4	60	49	
125	F 65-220/004 LG-125	ULG	1,23	0,80	2,75	17,4	55	49	39100099
	F 65-220/004 LG-125	YLG	1,23	0,80	2,75	17,4	40	49	39100100
	F 65-220/004 LG-125	WLG	1,23	0,80	2,75	17,4	60	49	
135	F 65-220/004 LG-135	ULG	1,23	0,80	2,75	17,4	55	49	39100101
	F 65-220/004 LG-135	YLG	1,23	0,80	2,75	17,4	40	49	39100102
	F 65-220/004 LG-135	WLG	1,23	0,80	2,75	17,4	60	49	
145	F 65-220/004 LG-145	ULG	1,23	0,80	2,75	17,4	55	49	39100103
	F 65-220/004 LG-145	YLG	1,23	0,80	2,75	17,4	40	49	39100104
	F 65-220/004 LG-145	WLG	1,23	0,80	2,75	17,4	60	49	
155	F 65-220/004 LG-155	ULG	1,23	0,80	2,75	17,4	55	49	39100105
	F 65-220/004 LG-155	YLG	1,23	0,80	2,75	17,4	40	49	39100106
	F 65-220/004 LG-155	WLG	1,23	0,80	2,75	17,4	60	49	
165	F 65-220/014 LG-165	ULG	1,94	1,30	3,54	17,4	55	50	39100107
	F 65-220/014 LG-165	YLG	1,94	1,30	3,54	17,4	40	50	39100108
	F 65-220/014 LG-165	WLG	1,94	1,30	3,54	17,4	60	50	
175	F 65-220/014 LG-175	ULG	1,94	1,30	3,54	17,4	55	50	39100109
	F 65-220/014 LG-175	YLG	1,94	1,30	3,54	17,4	40	50	39100110
	F 65-220/014 LG-175	WLG	1,94	1,30	3,54	17,4	60	50	
185	F 65-220/024 LG-185	ULG	2,56	1,80	4,25	17,4	55	51	39100111
	F 65-220/024 LG-185	YLG	2,56	1,80	4,25	17,4	40	51	39100112
	F 65-220/024 LG-185	WLG	2,56	1,80	4,25	17,4	60	51	
195	F 65-220/024 LG-195	ULG	2,56	1,80	4,25	17,4	55	51	39100113
	F 65-220/024 LG-195	YLG	2,56	1,80	4,25	17,4	40	51	39100114
	F 65-220/024 LG-195	WLG	2,56	1,80	4,25	17,4	60	51	
120	F 80-220/034 LG-120	ULG	2,60	1,90	5,87	37,5	55	63	39100123
	F 80-220/034 LG-120	YLG	2,60	1,90	5,87	37,5	40	63	39100124
	F 80-220/034 LG-120	WLG	2,60	1,90	5,87	37,5	60	63	
135	F 80-220/034 LG-135	ULG	2,60	1,90	5,87	37,5	55	63	39100137
	F 80-220/034 LG-135	YLG	2,60	1,90	5,87	37,5	40	63	39100138
	F 80-220/034 LG-135	WLG	2,60	1,90	5,87	37,5	60	63	
150	F 80-220/034 LG-150	ULG	2,60	1,90	5,87	37,5	55	63	39100139
	F 80-220/034 LG-150	YLG	2,60	1,90	5,87	37,5	40	63	39100140
	F 80-220/034 LG-150	WLG	2,60	1,90	5,87	37,5	60	63	
165	F 80-220/034 LG-165	ULG	3,50	2,60	6,50	37,5	55	63	39100129
	F 80-220/034 LG-165	YLG	3,50	2,60	6,50	37,5	40	63	39100130
	F 80-220/034 LG-165	WLG	3,50	2,60	6,50	37,5	60	64	
180	F 80-220/044 LG-180	ULG	5,13	3,70	8,40	37,5	55	65	39100131
	F 80-220/044 LG-180	YLG	5,13	3,70	8,40	37,5	40	65	39100132
	F 80-220/044 LG-180	WLG	5,13	3,70	8,40	37,5	60	66	
195	F 80-220/044 LG-195	ULG	5,13	3,70	8,40	37,5	55	65	39100133
	F 80-220/044 LG-195	YLG	5,13	3,70	8,40	37,5	40	65	39100134
	F 80-220/044 LG-195	WLG	5,13	3,70	8,40	37,5	60	66	
210	F 80-220/044 LG-210	ULG	5,13	3,70	8,40	37,5	55	65	39100135
	F 80-220/044 LG-210	YLG	5,13	3,70	8,40	37,5	40	65	39100136
	F 80-220/044 LG-210	WLG	5,13	3,70	8,40	37,5	60	66	
120	F 100-220/034 LG-120	ULG	2,60	1,90	5,87	37,5	55	64	39100145
	F 100-220/034 LG-120	YLG	2,60	1,90	5,87	37,5	40	64	39100146
	F 100-220/034 LG-120	WLG	2,60	1,90	5,87	37,5	60	64	

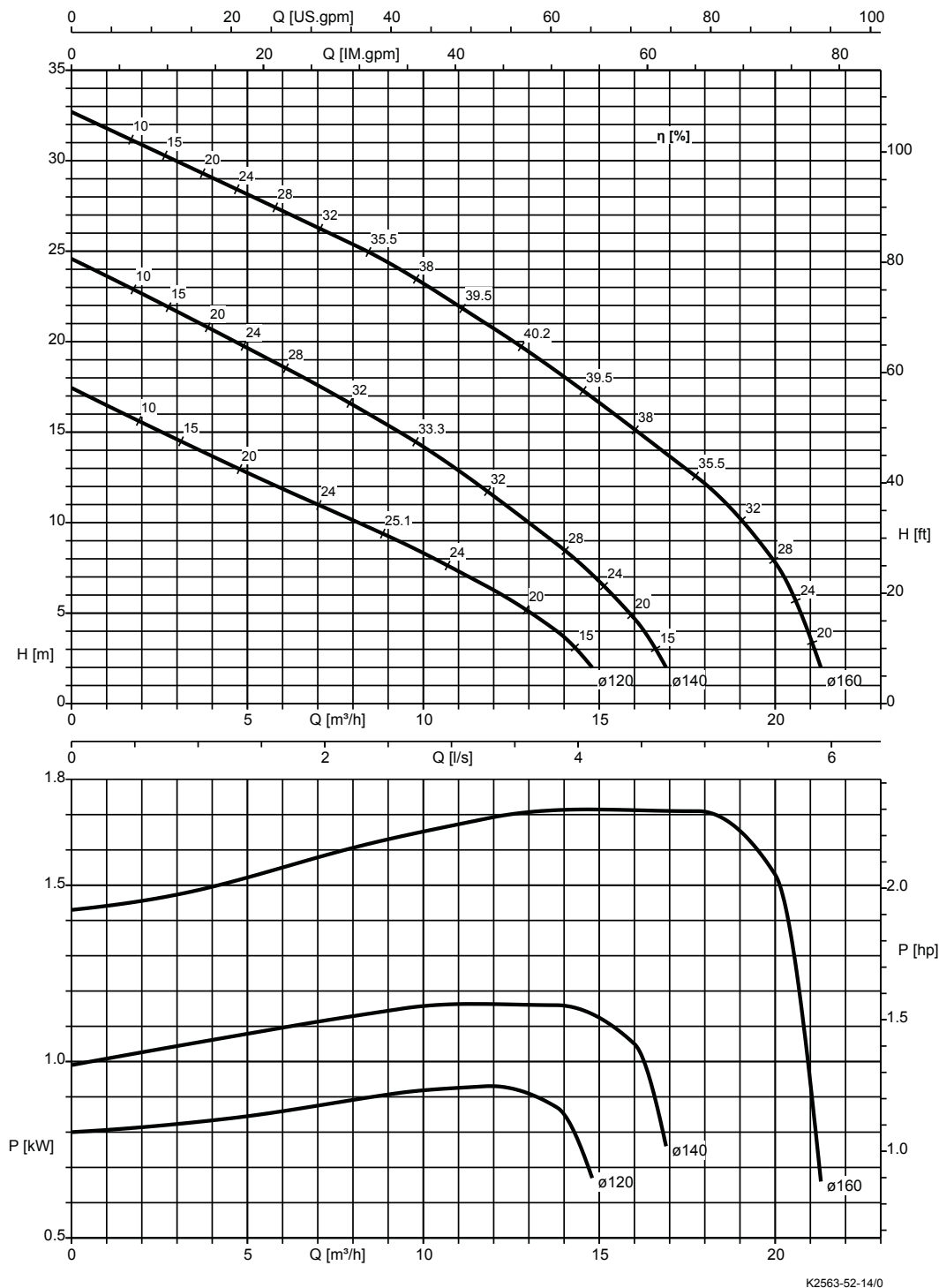
Impeller diameter	Size	Motor version	P ₁	P ₂	I _N	I _A	Fluid temperature	[kg]	Mat. No.
			[kW]	[kW]	[A]	[A]	[°C]		
135	F 100-220/034 LG-135	ULG	2,60	1,90	5,87	37,5	55	64	39100159
	F 100-220/034 LG-135	YLG	2,60	1,90	5,87	37,5	40	64	39100160
	F 100-220/034 LG-135	WLG	2,60	1,90	5,87	37,5	60	64	
150	F 100-220/034 LG-150	ULG	3,50	2,60	6,50	37,5	55	64	39100149
	F 100-220/034 LG-150	YLG	3,50	2,60	6,50	37,5	40	64	39100150
	F 100-220/034 LG-150	WLG	3,50	2,60	6,50	37,5	60	64	
165	F 100-220/044 LG-165	ULG	5,13	3,70	8,40	37,5	55	65	39100151
	F 100-220/044 LG-165	YLG	5,13	3,70	8,40	37,5	40	65	39100152
	F 100-220/044 LG-165	WLG	5,13	3,70	8,40	37,5	60	67	
180	F 100-220/044 LG-180	ULG	5,13	3,70	8,40	37,5	55	66	39100153
	F 100-220/044 LG-180	YLG	5,13	3,70	8,40	37,5	40	66	39100154
	F 100-220/044 LG-180	WLG	5,13	3,70	8,40	37,5	60	67	
195	F 100-220/044 LG-195	ULG	5,13	3,70	8,40	37,5	55	67	39100155
	F 100-220/044 LG-195	YLG	5,13	3,70	8,40	37,5	40	67	39100156
	F 100-220/044 LG-195	WLG	5,13	3,70	8,40	37,5	60	67	
210	F 100-220/044 LG-210	ULG	5,13	3,70	8,40	37,5	55	67	39100157
	F 100-220/044 LG-210	YLG	5,13	3,70	8,40	37,5	40	67	39100158
	F 100-220/044 LG-210	WLG	5,13	3,70	8,40	37,5	60	67	
154	D 80-220/034 LG-154	ULG	2,60	1,90	5,87	37,5	55	74	39100345
	D 80-220/034 LG-154	YLG	2,60	1,90	5,87	37,5	40	74	39100346
	D 80-220/034 LG-154	WLG	2,60	1,90	5,87	37,5	60	75	
168	D 80-220/034 LG-168	ULG	2,60	1,90	5,87	37,5	55	74	39100347
	D 80-220/034 LG-168	YLG	2,60	1,90	5,87	37,5	40	74	39100348
	D 80-220/034 LG-168	WLG	2,60	1,90	5,87	37,5	60	75	
180	D 80-220/034 LG-180	ULG	2,60	1,90	5,87	37,5	55	74	39100349
	D 80-220/034 LG-180	YLG	2,60	1,90	5,87	37,5	40	74	39100350
	D 80-220/034 LG-180	WLG	2,60	1,90	5,87	37,5	60	75	
190	D 80-220/034 LG-190	ULG	2,60	1,90	5,87	37,5	55	75	39100351
	D 80-220/034 LG-190	YLG	2,60	1,90	5,87	37,5	40	75	39100352
	D 80-220/034 LG-190	WLG	2,60	1,90	5,87	37,5	60	75	
195	D 100-220/034 LG-195	ULG	3,50	2,60	6,50	37,5	55	79	39100366
	D 100-220/034 LG-195	YLG	3,50	2,60	6,50	37,5	40	79	39100367
	D 100-220/034 LG-195	WLG	3,50	2,60	6,50	37,5	60	79	
209	D 100-220/044 LG-209	ULG	5,13	3,70	8,40	37,5	55	79	39100368
	D 100-220/044 LG-209	YLG	5,13	3,70	8,40	37,5	40	79	39100369
	D 100-220/044 LG-209	WLG	5,13	3,70	8,40	37,5	60	80	
220	D 100-220/044 LG-220	ULG	5,13	3,70	8,40	37,5	55	80	39100370
	D 100-220/044 LG-220	YLG	5,13	3,70	8,40	37,5	40	80	39100371
	D 100-220/044 LG-220	WLG	5,13	3,70	8,40	37,5	60	80	

Characteristic curves

n = 2900 rpm

Amarex N S 50-172, n = 2900 rpm

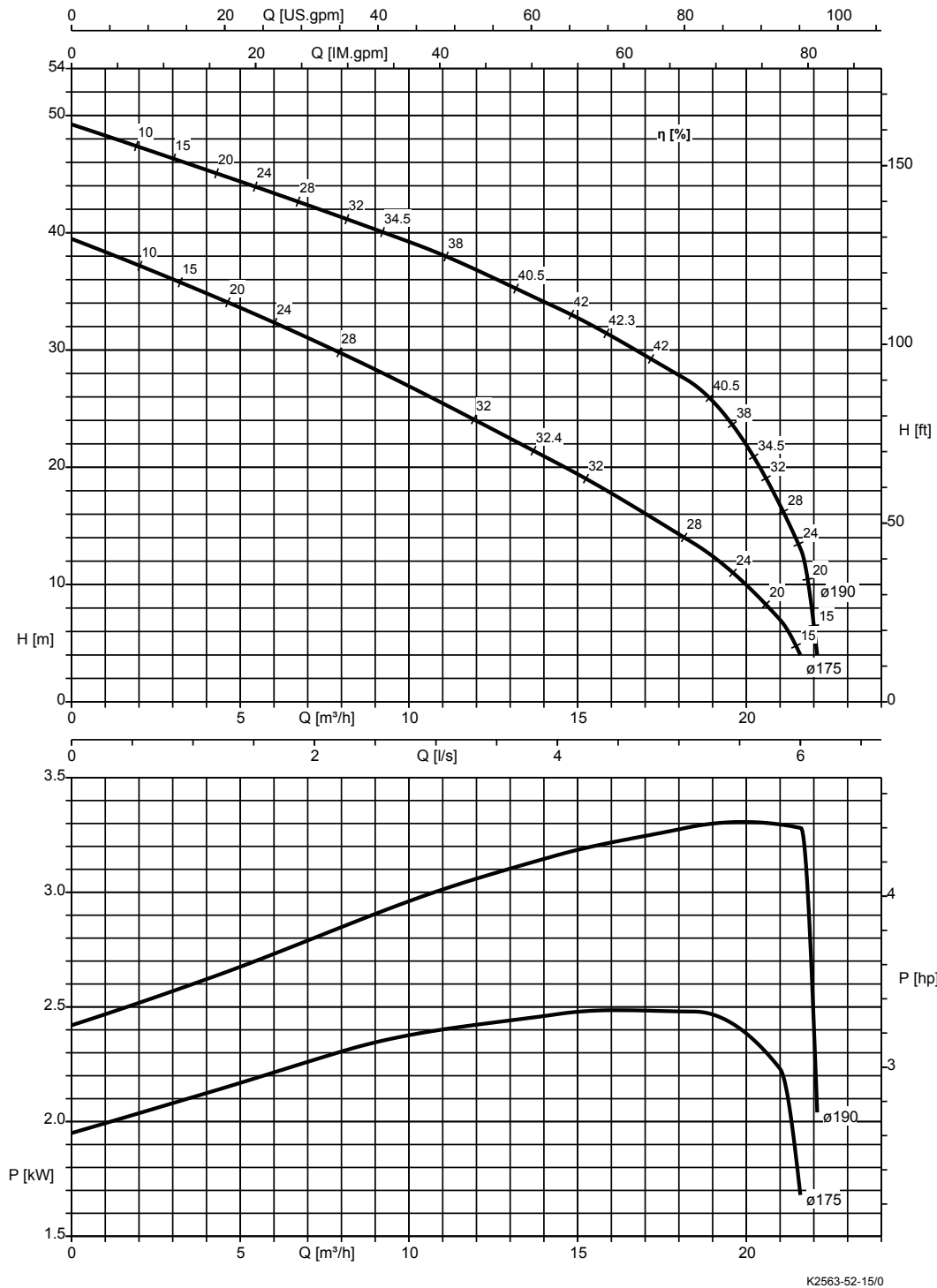
Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to § 4.4.2. The curves refer to the effective motor speed.



Free passage = 6 mm

Amarex N S 50-222, n = 2900 rpm

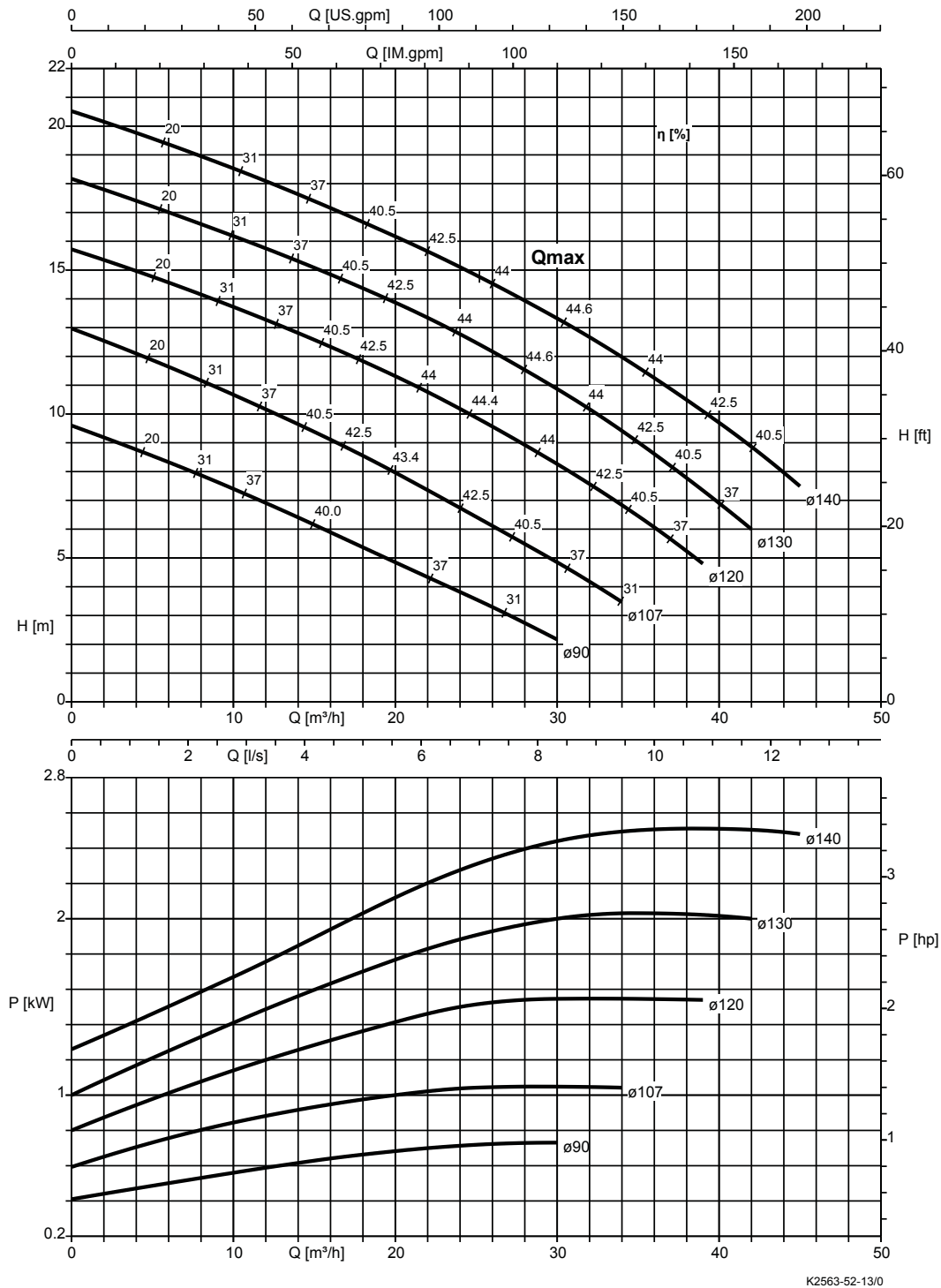
Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to
§ 4.4.2. The curves refer to the effective motor speed.



Free passage = 6 mm

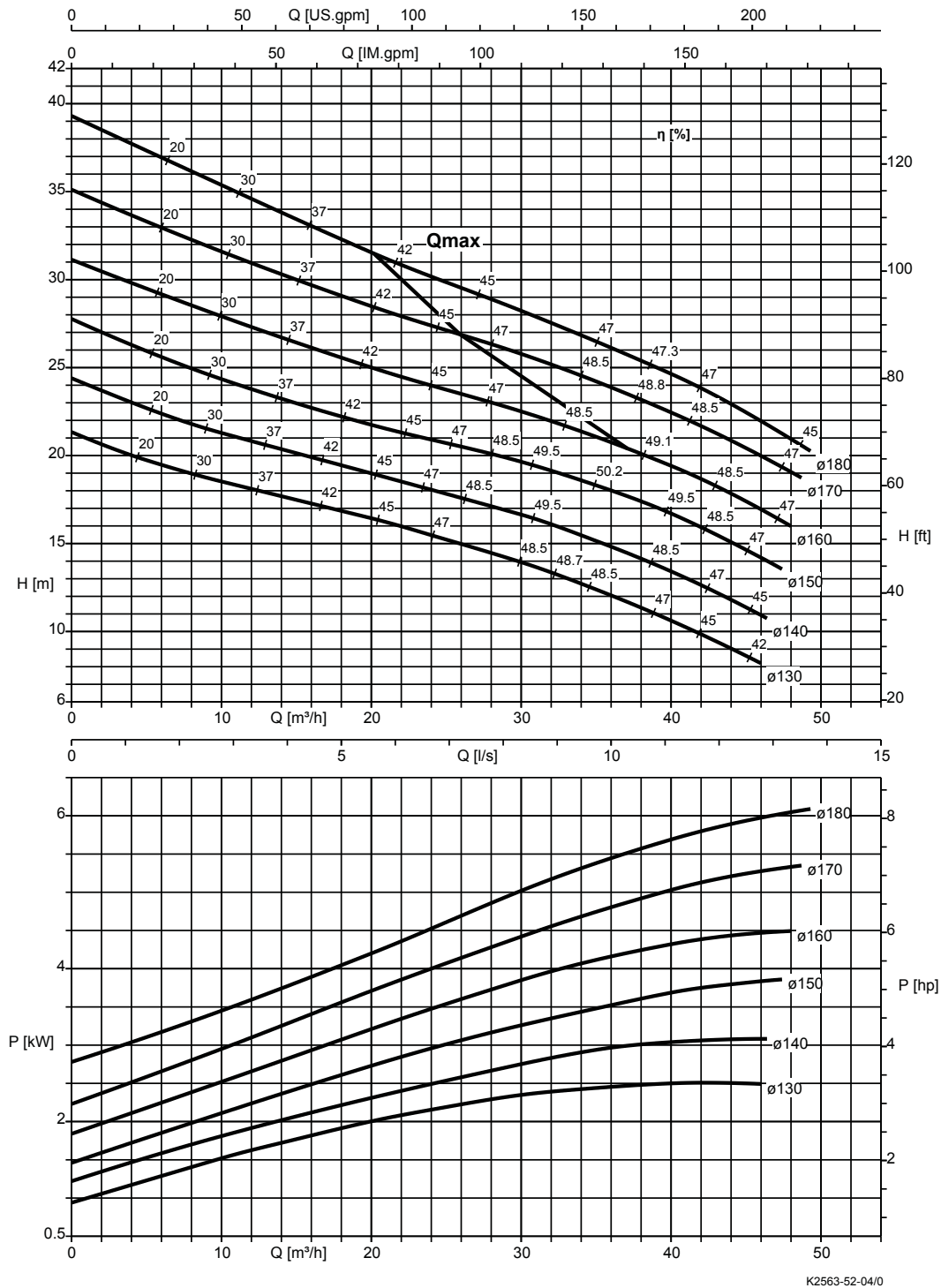
Amarex N F 50-170, n = 2900 rpm

Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to § 4.4.2. The curves refer to the effective motor speed.



Amarex N F 50-220, n = 2900 rpm

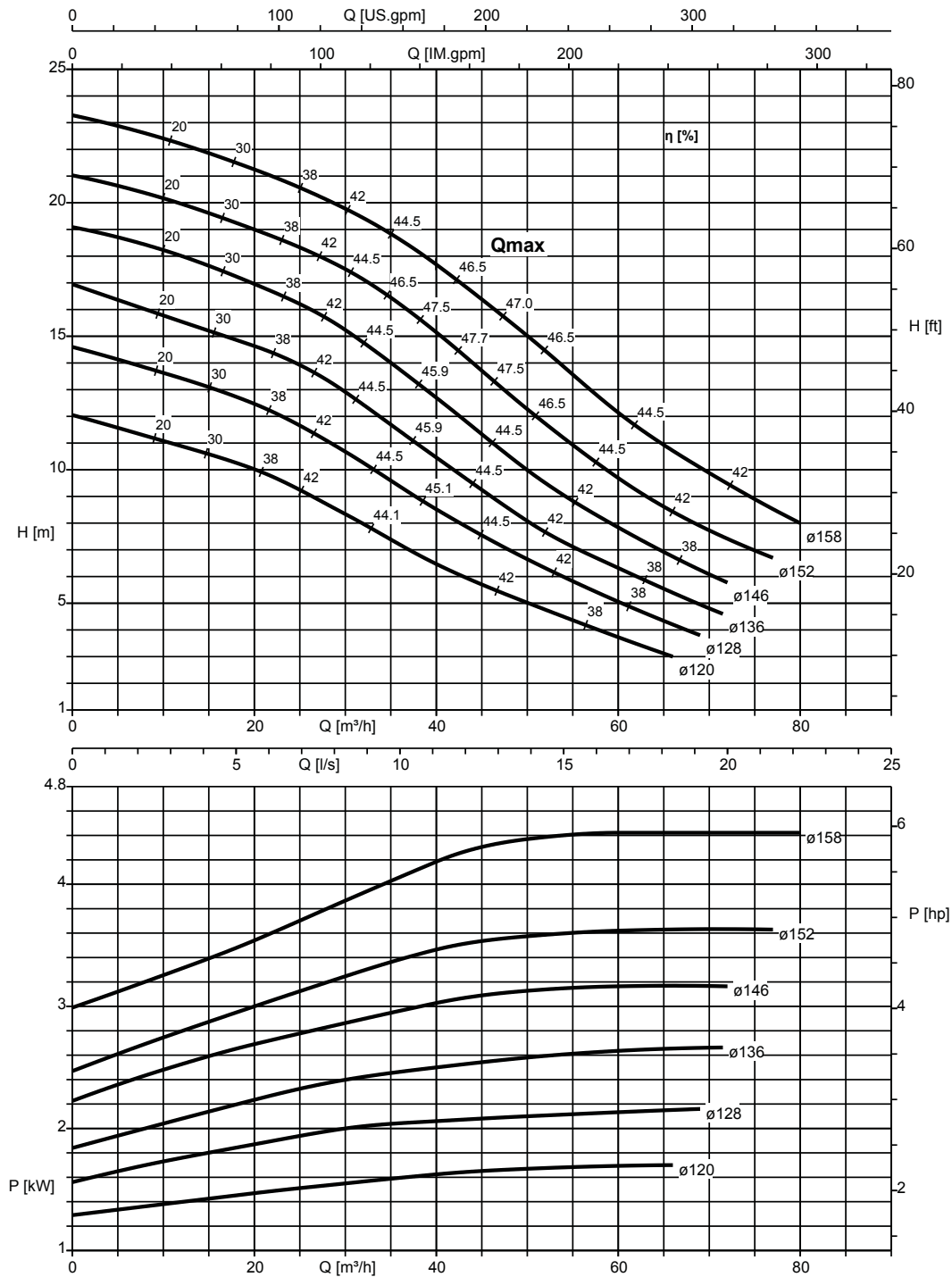
Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to § 4.4.2. The curves refer to the effective motor speed.



Free passage = 40 mm

Amarex N F 65-170, n = 2900 rpm

Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to
§ 4.4.2. The curves refer to the effective motor speed.

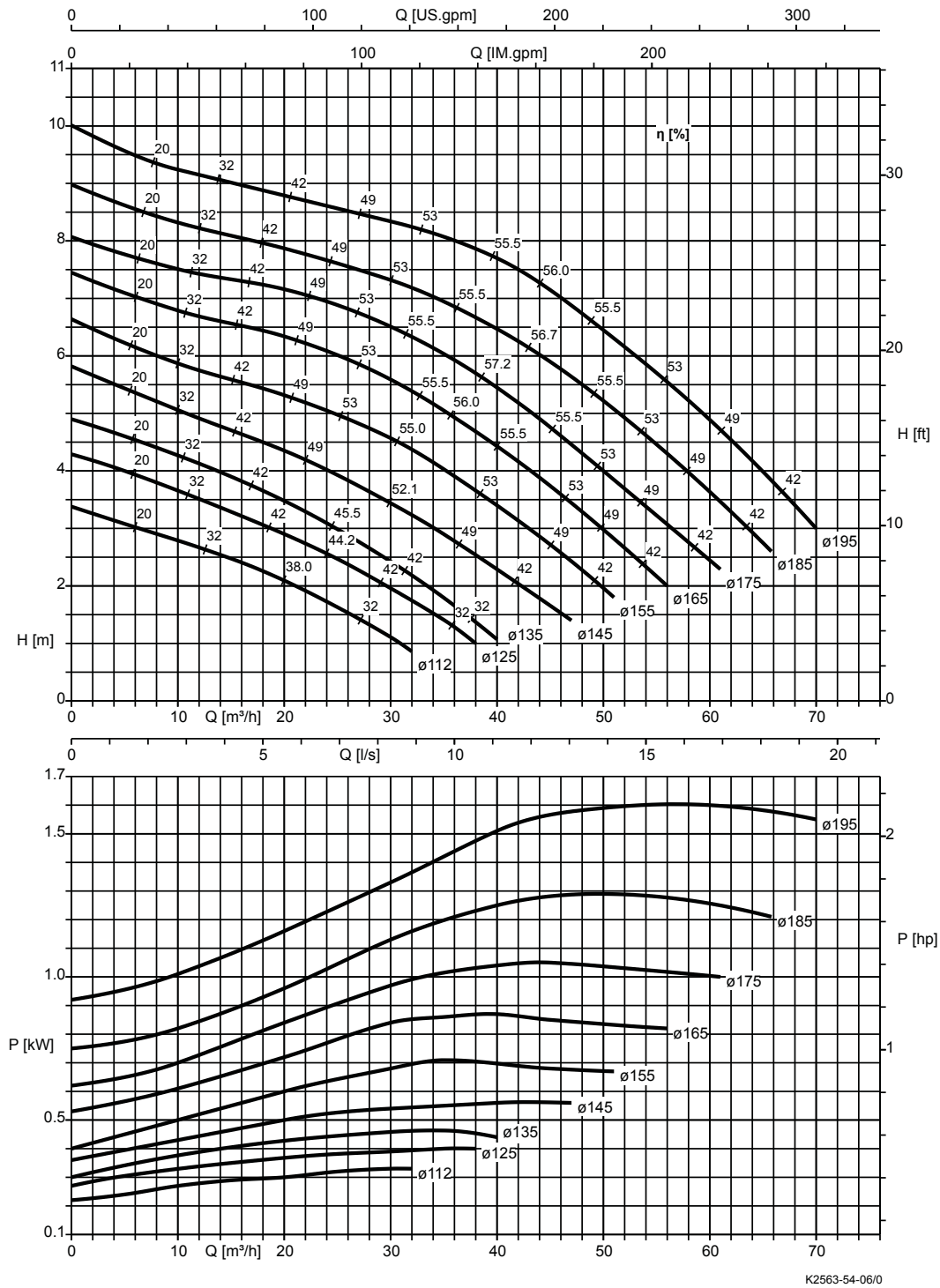


Free passage = 65 mm

n = 1450 rpm

Amarex N F 65-220, n = 1450 rpm

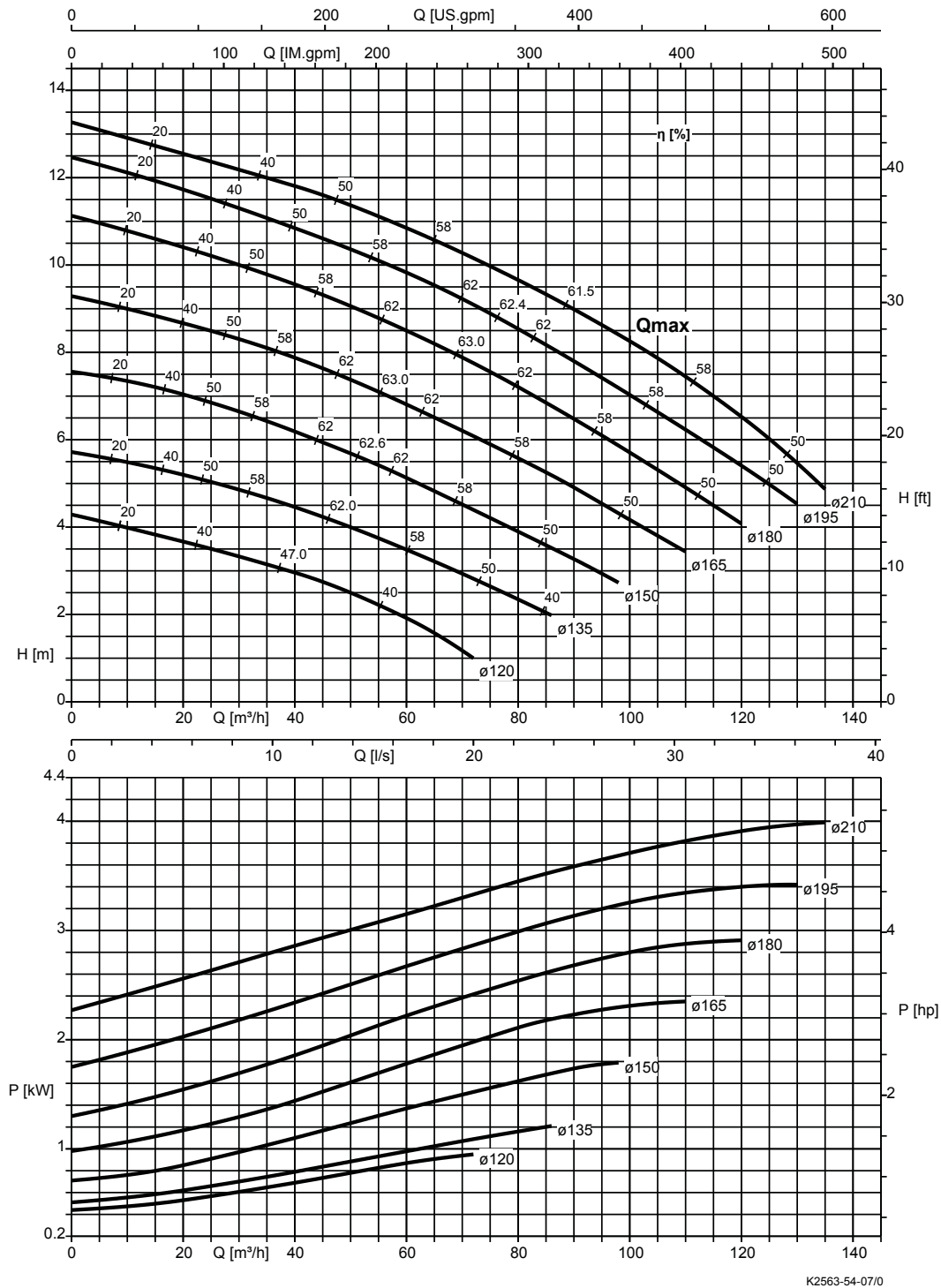
Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to § 4.4.2. The curves refer to the effective motor speed.



Free passage = 65 mm

Amarex N F 80-220, n = 1450 rpm

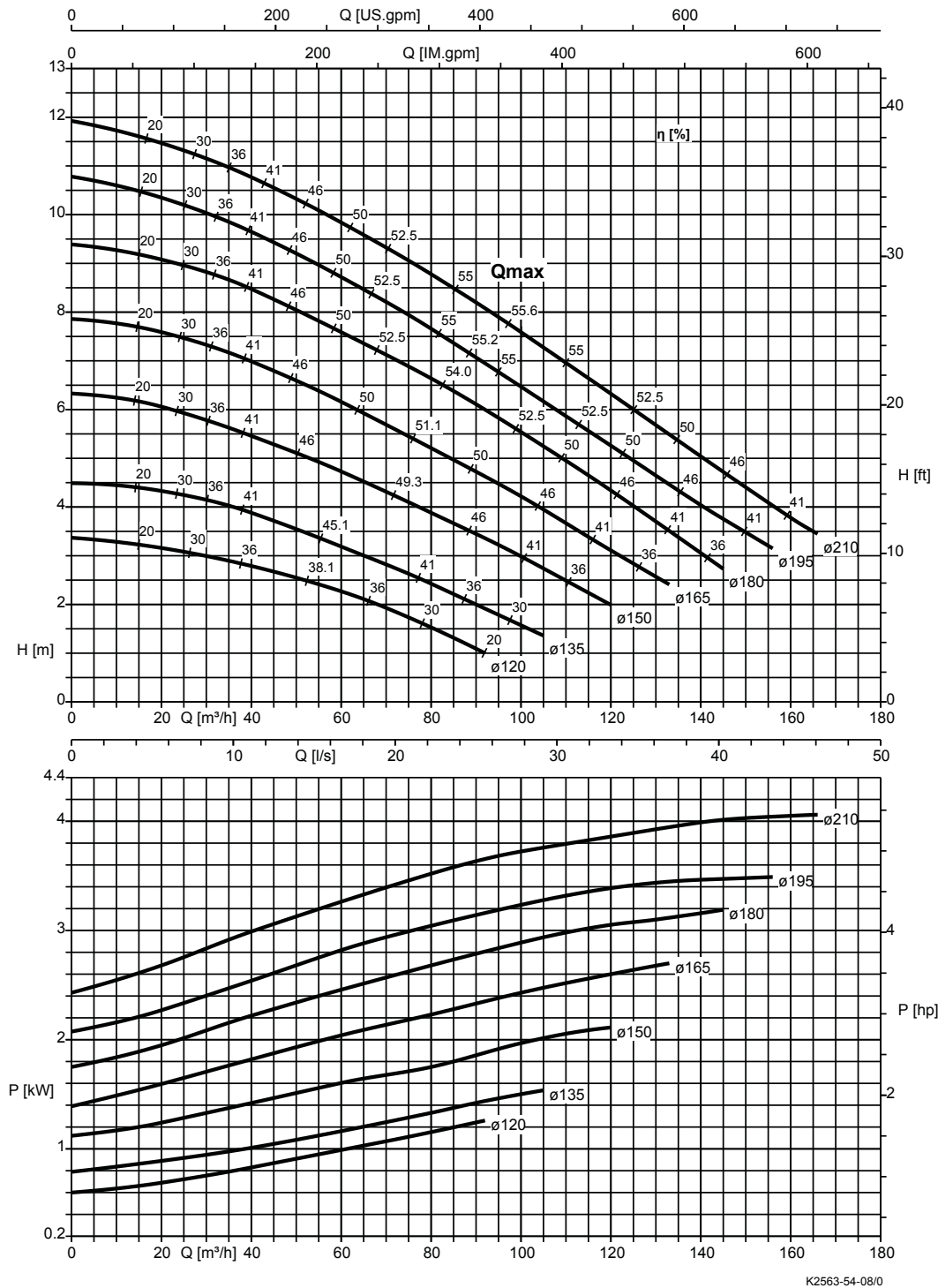
Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to § 4.4.2. The curves refer to the effective motor speed.



Free passage = 76 mm

Amarex N F 100-220, n = 1450 rpm

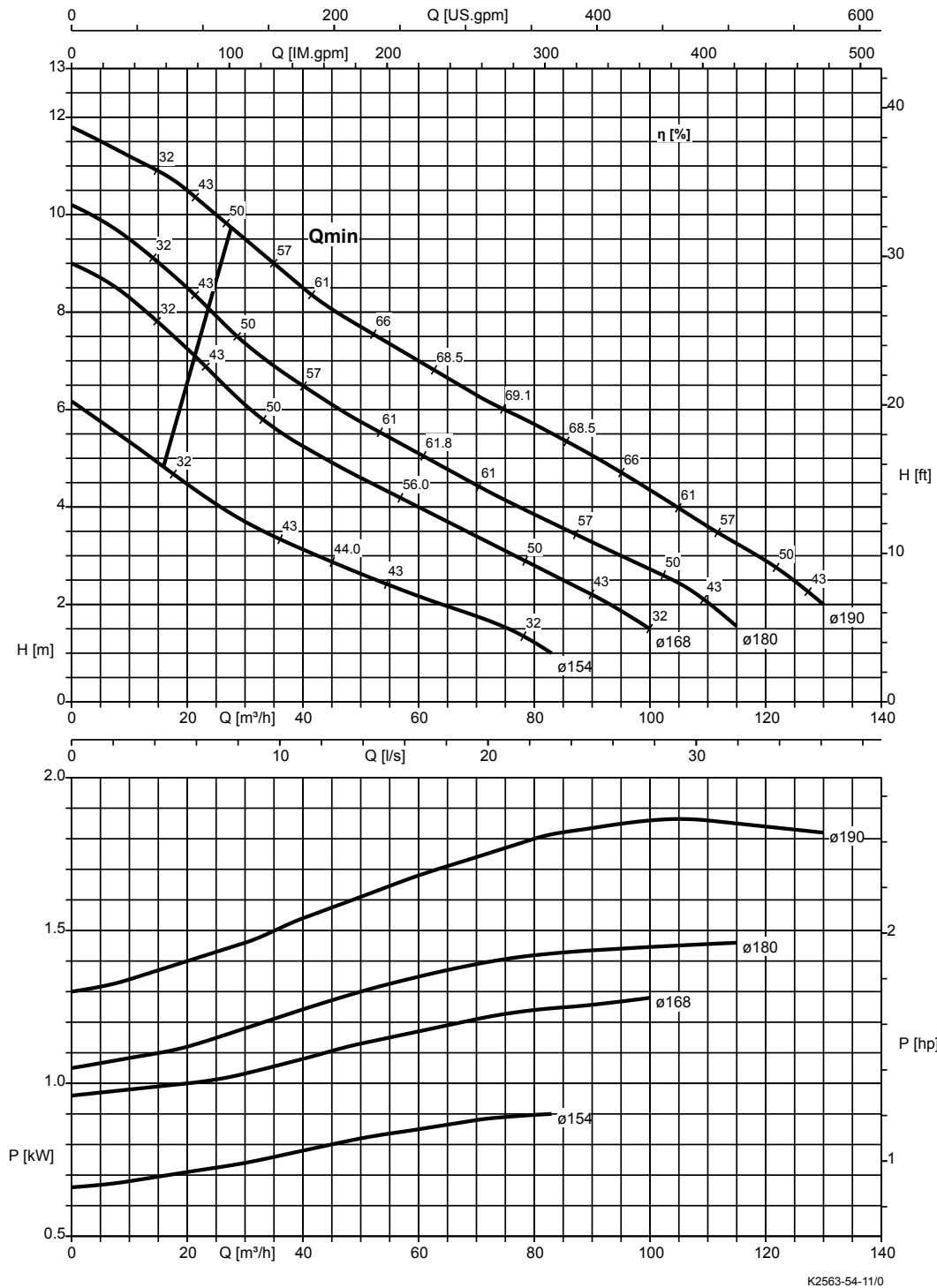
Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to
§ 4.4.2. The curves refer to the effective motor speed.



Free passage = 100 mm

Amarex N D 80-220, n = 1450 rpm

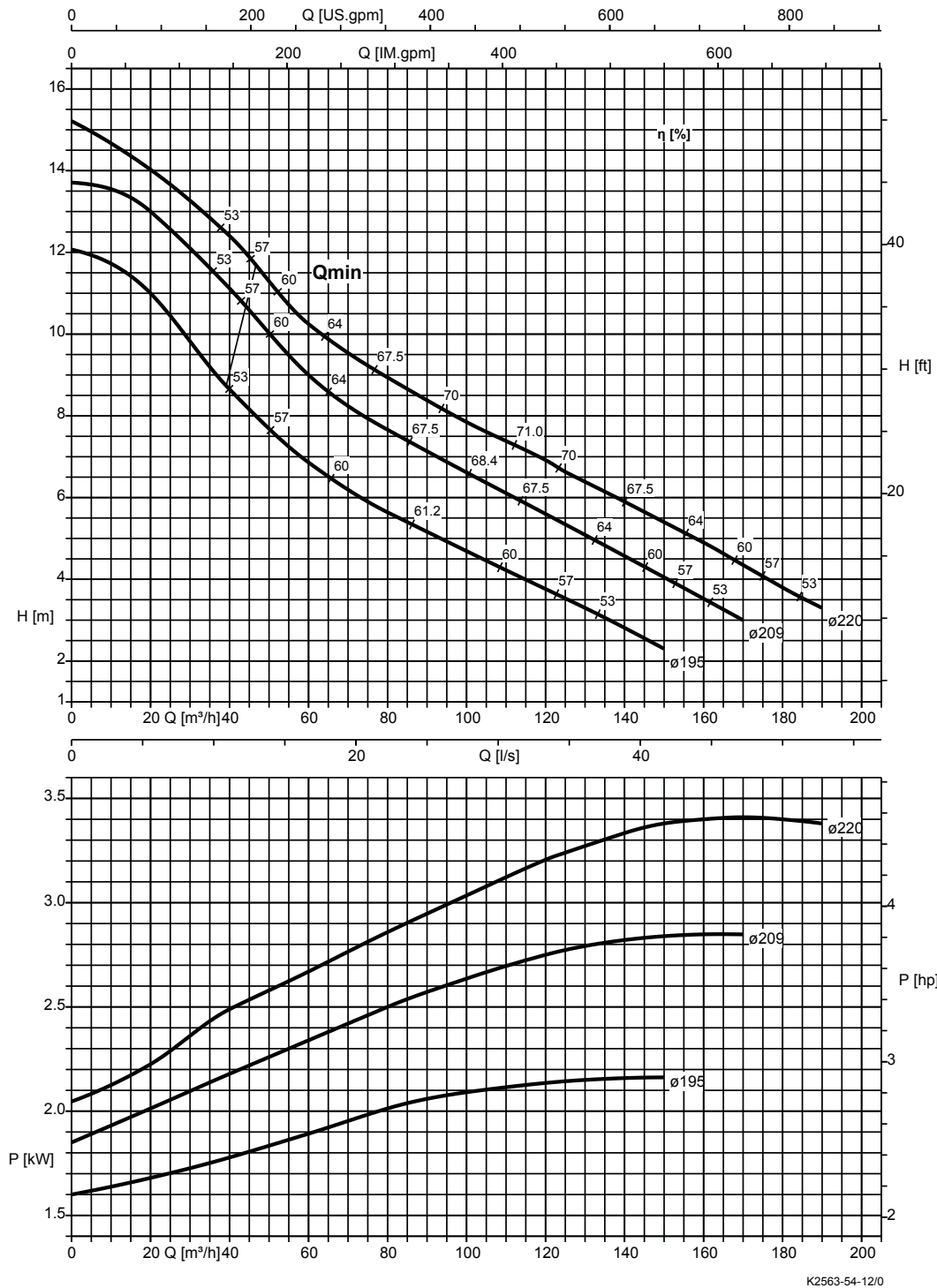
Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to § 4.4.2. The curves refer to the effective motor speed.



Free passage = 65 mm

Amarex N D 100-220, n = 1450 rpm

Characteristic curves to ISO 9906 Class 2A / 3B, below 10 kW to § 4.4.2. The curves refer to the effective motor speed.

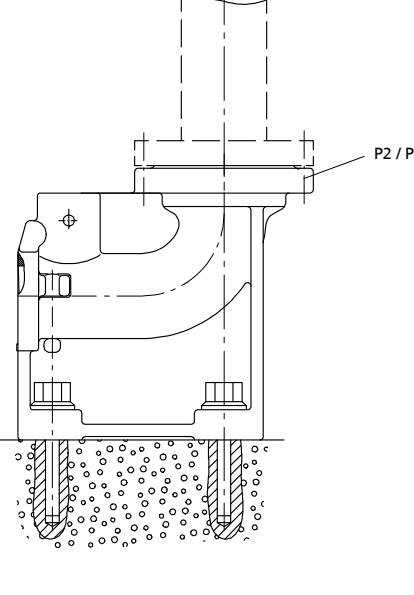
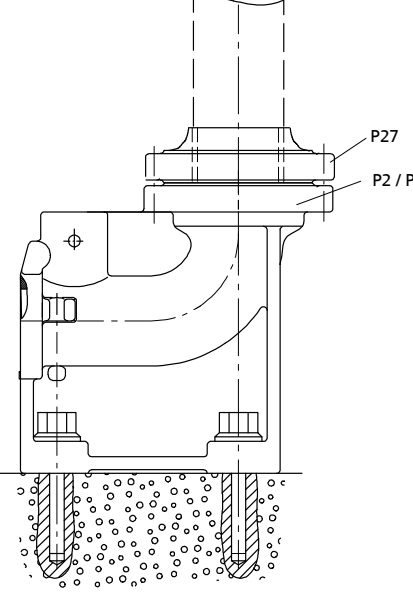
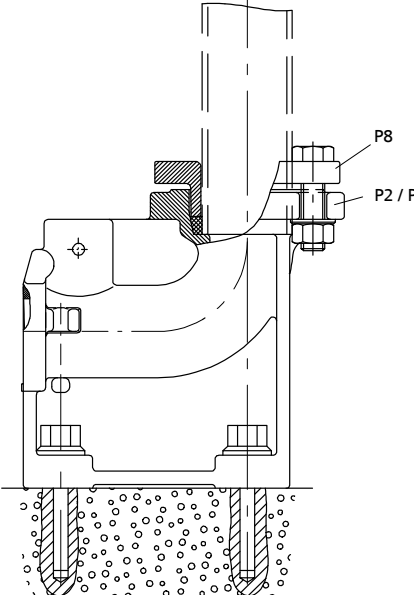


Free passage = 76 mm

Dimensions and connections

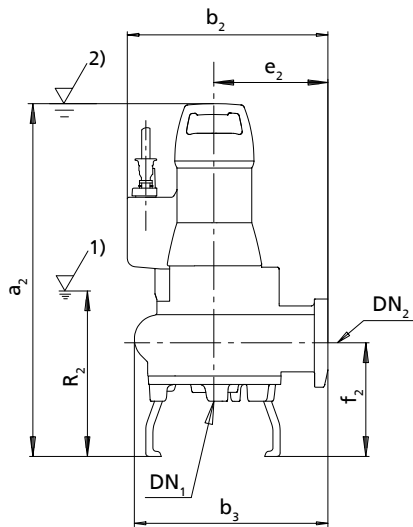
Connections

Discharge line connections at the duckfoot bend

Flanged connection (DN 50/DN 65)	Threaded connection in the flange (DN 50/DN 65)	Clamped connection (DN 50/DN 65)
		
	For standard pipes to DIN 2440 / DIN 2441 ▪ With outside diameter of 60.3 mm, steel for DN 50 ▪ With outside diameter of 63 mm, PVC (ISO 3606) for DN 50 ▪ With threaded flange DN 50 - G 2" ▪ With outside diameter of 76.1 mm, steel for DN 65 ▪ With outside diameter of 75 mm, PVC (ISO 3606) for DN 65 ▪ With threaded flange DN 65 - G 2 1/2"	For standard pipes to DIN 2440 / DIN 2441 / DIN 2448 ▪ With outside diameter of 60.3 mm, steel for DN 50 ▪ With outside diameter of 63 mm, PVC (ISO 3606) for DN 50 ▪ With outside diameter of 76.1 mm, steel for DN 65 ▪ With outside diameter of 75 mm, PVC (ISO 3606) for DN 65

Dimensions

Amarex N, transportable model

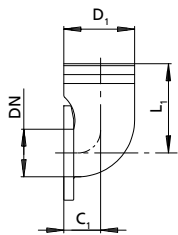


1)	Lowest switch-off point for automatic operation
2)	Minimum submergence for continuous operation

Pump dimensions [mm]

Size	Pump							
	DN ₁	DN ₂	a ₂ ¹⁸⁾	b ₂	b ₃	e ₂	f ₂ ¹⁸⁾	R ₂
50-172 S	-	50	547	322	293	180	152	207
50-170 F	50	50	547	322	293	180	152	207
50-222 S	-	50	609	336	307	180	155	203
50-220 F	50	50	609	336	307	180	155	203
65-170 F	65	65	653	367	338	210	164	248
65-220 F	65	65	593	353	347	210	163	253
80-220 F	80	80	672	386	392	230	187	249
80-220 D	-	80	672	386	392	230	187	249
100-220 F	100	100	698	383	390	230	207	277
100-220 D	-	100	698	383	390	230	207	277

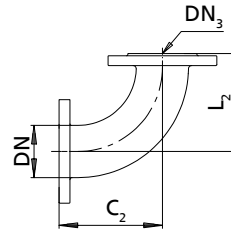
Connection elbow with flange/hose connection (P13)



Dimensions [mm]

DN	D ₁	C ₁	L ₁
65	75	40	135
80	75	115	175
100	110	45	195

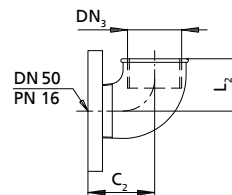
Connection elbow with flanges (P14)



Dimensions [mm]

DN	DN ₃ ¹⁹⁾	C ₂	L ₂
65	65	135	135
80	80	135	135
100	100	120	175

Connection elbow with internal and external thread (P14) and threaded flange (P27)

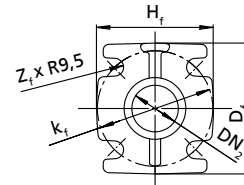


Dimensions [mm]

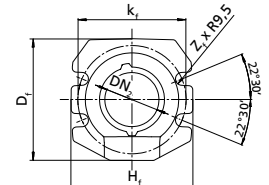
DN	DN ₃	C ₂	L ₂
50	G 2"	78	58

Pump flange DN₂

DN 50 and 65
 ISO 7005 PN 16
 DIN 2501 PN 16



DN 80 and 100
 ISO 7005 PN16
 DIN 2501 PN16



Dimensions of the pump flange in [mm]

Pump size	Flange			
	H _f	k _f	D _f	Z _f
50-172 S	125	125	140	4
50-170 F	125	125	140	4
50-222 S	125	125	140	4
50-220 F	125	125	140	4
65-170 F	144	145	164	4
65-220 F	144	145	164	4
80-220 F	180	160	180	4
80-220 D	180	160	180	4
100-220 F	202	180	205	4
100-220 D	202	180	205	4

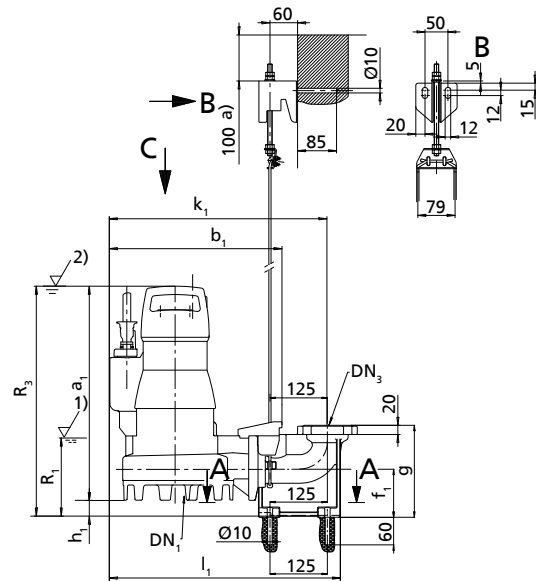
¹⁸⁾ If a foot plate is fitted, a₂ + 10 mm

¹⁹⁾ DN₃ to ISO 7005/DIN 2501

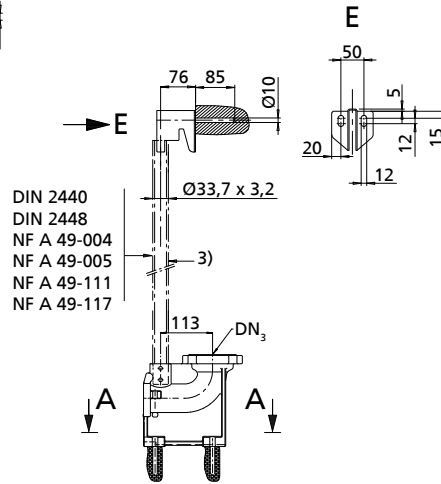
Amarex N 50, stationary model with guide wire, single-rail and guide hoop arrangement, straight claw

DN 3 = DN 50: DIN ISO ASME = standard

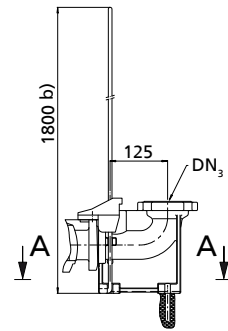
Guide wire arrangement



Single guide rail arrangement



Guide hoop arrangement

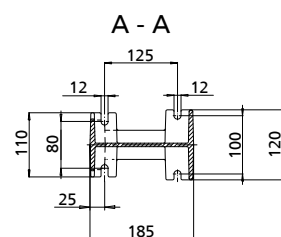


1)	Lowest switch-off point for automatic operation	2)	Minimum submergence for continuous operation
3)	Not included in KSB's scope of supply	a)	Minimum
b)	Maximum		

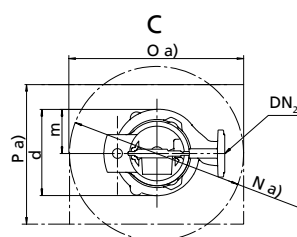
Dimensions of pump and foundation [mm]

Size	Pump															Foundation			
	DN ₁	DN ₂	a ₁	b ₁	d	f ₁	g	h ₁	k ₁	l ₁	m	R ₁	R ₃	Z _f	DN ₃	N	O	P	
50–172 S	-	50	470	376	250	105	200	31	472	502	125	161	501	4	50	465	465	350	
50–170 F	50	50	470	376	250	105	200	31	472	502	125	161	501	4	50	465	465	350	
50–222 S	-	50	532	389	254	105	200	27	488	514	129	153	559	4	50	465	465	350	
50–220 F	50	50	532	389	254	105	200	27	488	514	129	153	559	4	50	465	465	350	

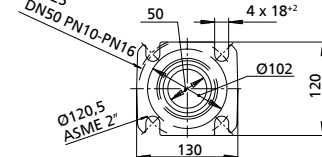
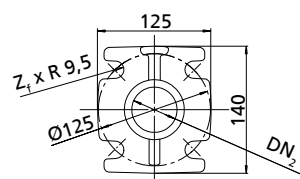
Installation in the sump



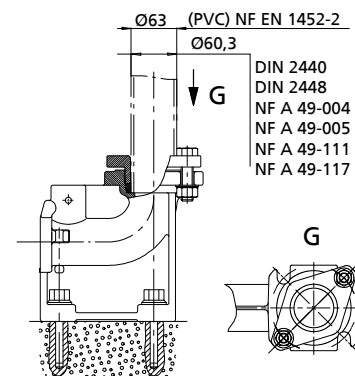
Pump flange DN₂
ISO 7005 PN10/16
DIN 2501 PN10/16



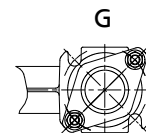
Flanged elbow DN₃
ISO 7005 PN 10 - PN16
DIN 2501 PN10 - PN16
ASME 2"
ASME B16.1 class 125



Clamped connection



DIN 2440
DIN 2448
NF A 49-004
NF A 49-005
NF A 49-111
NF A 49-117



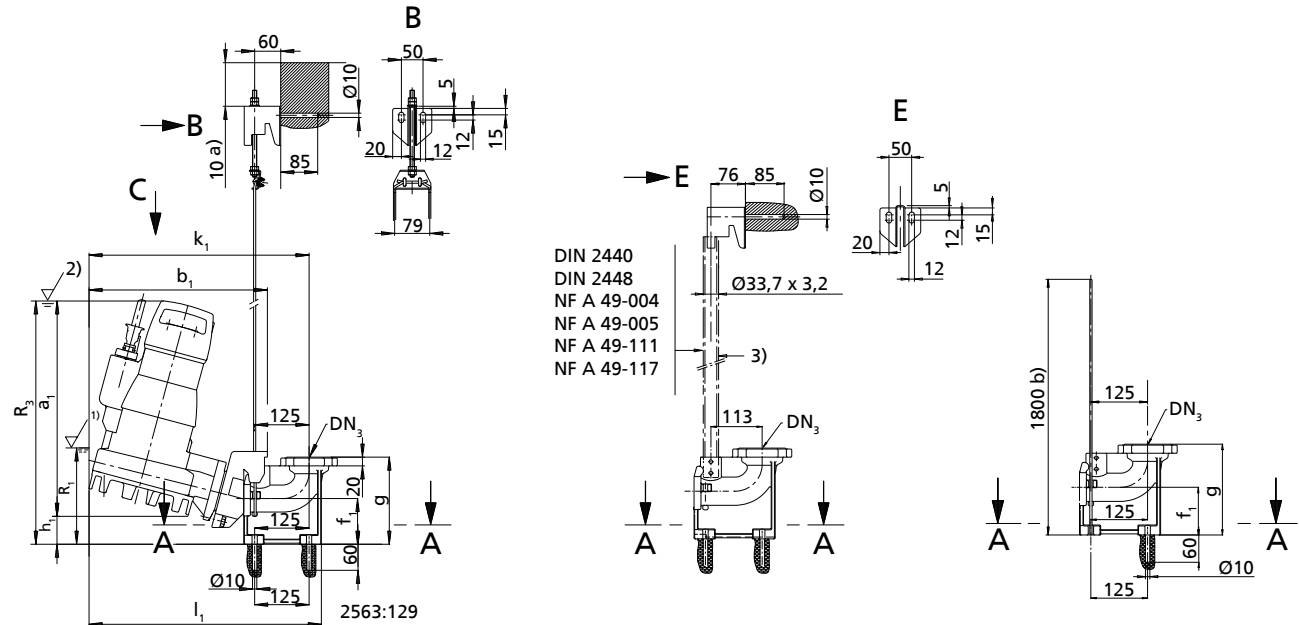
Amarex N 50, stationary model with guide wire, single-rail and guide hoop arrangement, inclined claw

DN 3 = DN 50: DIN ISO ASME = standard

Guide wire arrangement

Single guide rail arrangement

Guide hoop arrangement



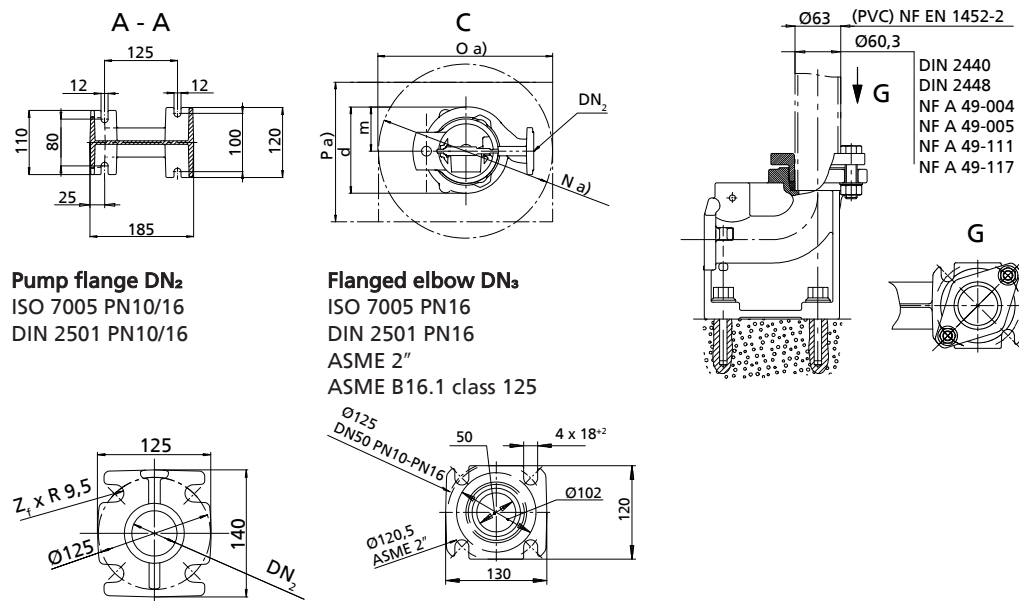
1)	Lowest switch-off point for automatic operation	2)	Minimum submergence for continuous operation
3)	Not included in KSB's scope of supply	a)	Minimum
b)	Maximum		

Dimensions of pump and foundation [mm]

Size	Pump													Foundation			
	DN ₂	a ₁	b ₁	d	f ₁	g	h ₁	k ₁	l ₁	m	R ₁	R ₃	Z _f	DN ₃	N	O	P
50-172 S	50	495	421	250	105	200	58	500	526	125	220	550	4	50	480	480	350
50-222 S	50	556	416	254	105	200	54	506	532	129	230	606	4	50	480	480	350

Installation in the sump

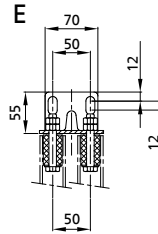
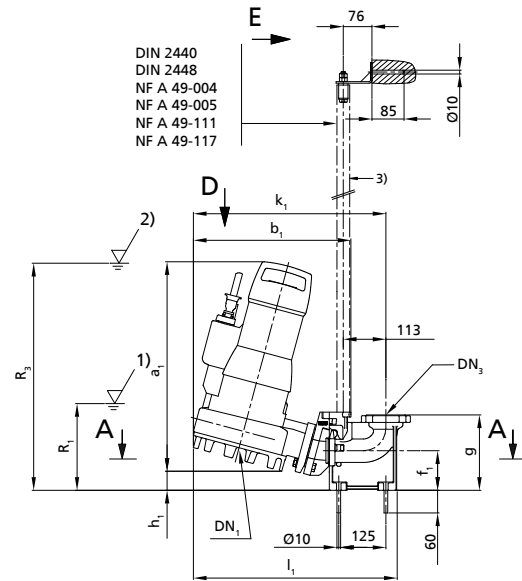
Clamped connection



Amarex N 50, stationary installation with twin guide rail arrangement, inclined claw

DN 3 = DN 50: DIN ISO ASME = standard

Twin guide rail arrangement

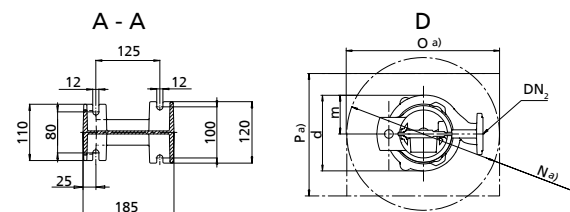


1)	Lowest switch-off point for automatic operation	2)	Minimum submergence for continuous operation
3)	Not included in KSB's scope of supply	a)	Minimum

Dimensions of pump and foundation [mm]

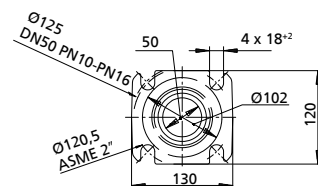
Size	Pump														Foundation			
	DN ₁	DN ₂	a ₁	b ₁	d	f ₁	g	h ₁	k ₁	l ₁	m	R ₁	R ₃	Z _f	DN ₃	N	O	P
50-172 S	-	50	494	422	250	105	200	54	499	528	125	220	550	4	50	480	480	350
50-170 F	50	50	494	422	250	105	200	54	499	528	125	220	550	4	50	480	480	350
50-222 S	-	50	549	426	254	105	200	53	506	535	129	230	606	4	50	480	480	350
50-220 F	50	50	549	426	254	105	200	53	506	535	129	230	606	4	50	480	480	350

Installation in the sump

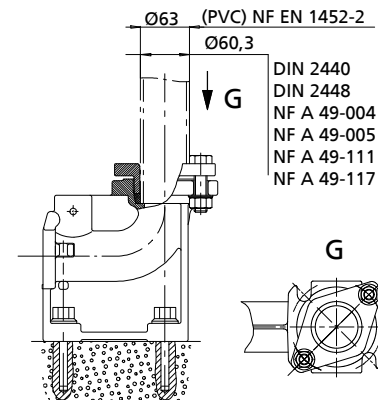


Flanged elbow DN₃

ISO 7005 PN16
DIN 2501 PN16
ASME 2"
ASME B16.1 class 125



Clamped connection



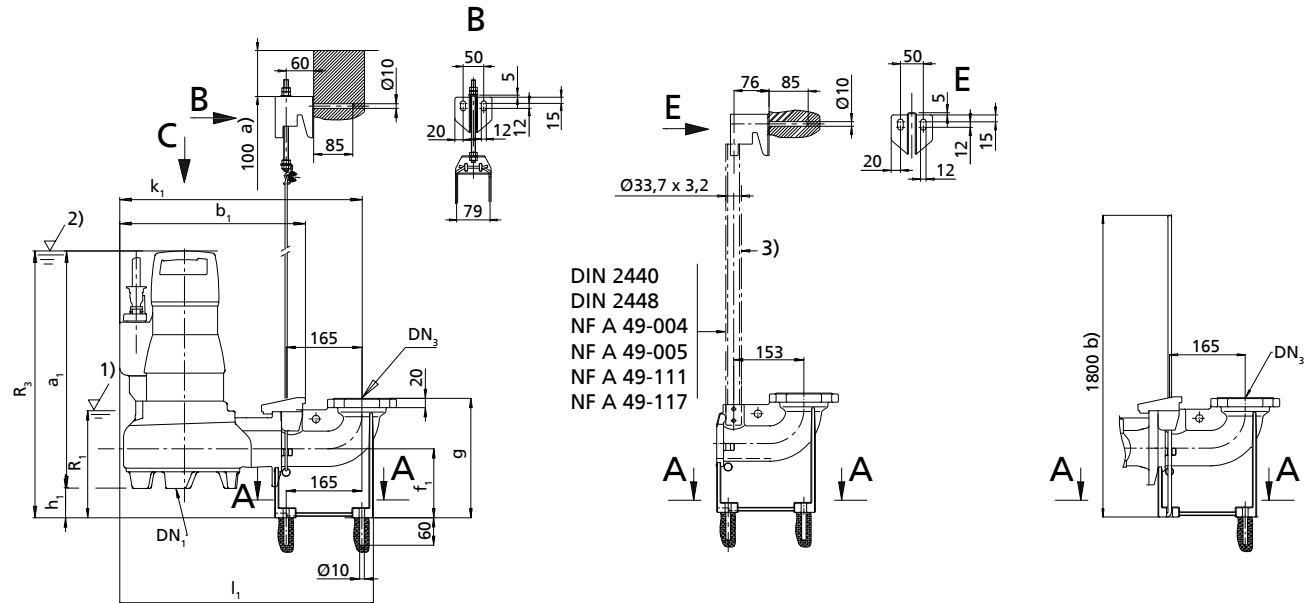
Amarex N 65, stationary model with guide wire, single-rail and guide hoop arrangement

DN 3 = 65/65: DIN ISO ASME = standard - DN 3 = 65/80: DIN ISO = standard, ASME = variant

Guide wire arrangement

Single guide rail arrangement

Guide hoop arrangement

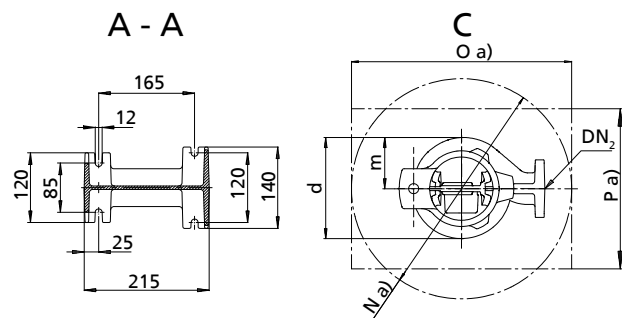


1)	Lowest switch-off point for automatic operation	2)	Minimum submergence for continuous operation
3)	Not included in KSB's scope of supply	a)	Minimum
b)	Maximum		

Dimensions of pump and foundation [mm]

Size	Pump														Foundation			
	DN ₁	DN ₂	a ₁	b ₁	d	f ₁	g	h ₁	k ₁	l ₁	m	R ₁	R ₃	Z _f	DN ₃	N	O	P
65-170 F	65	65	578	422	251	150	260	61	558	583	127	234	639	4	65	500	500	400
65-220 F	65	65	518	407	265	150	260	63	544	569	142	241	581	4	65	500	500	400

Installation in the sump



Flanged elbow DN

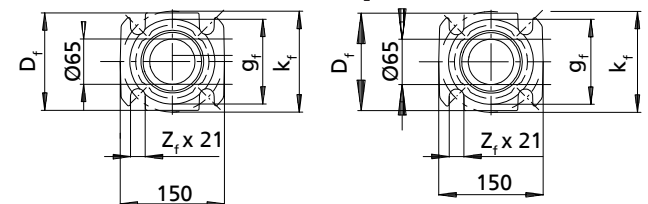
DN 65/65

ISO 7005 PN16

DIN 2501 PN16

ASME B 16.1 class 125

2 1/2"



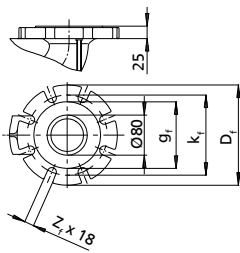
Dimensions of the flanged elbow DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	65	122	145	140	4
DIN 2501 PN16	80	133	160	200	8
ASME B16.1 class 125	65	122	140	140	4
	80	127	152,5	191	4

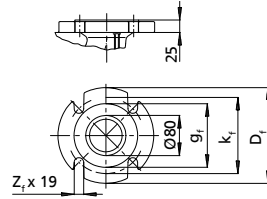
Extension elbow DN₃

DN 65/80

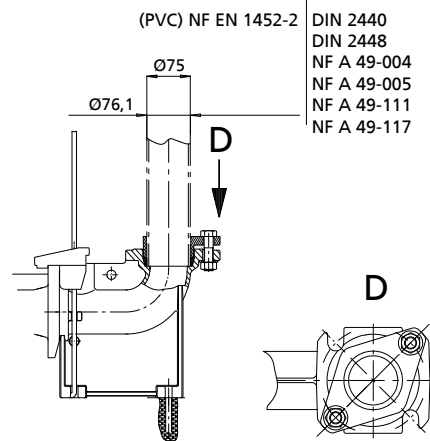
ISO 7005 PN16
DIN 2501 PN16



ASME B 16.1 class 125
3"



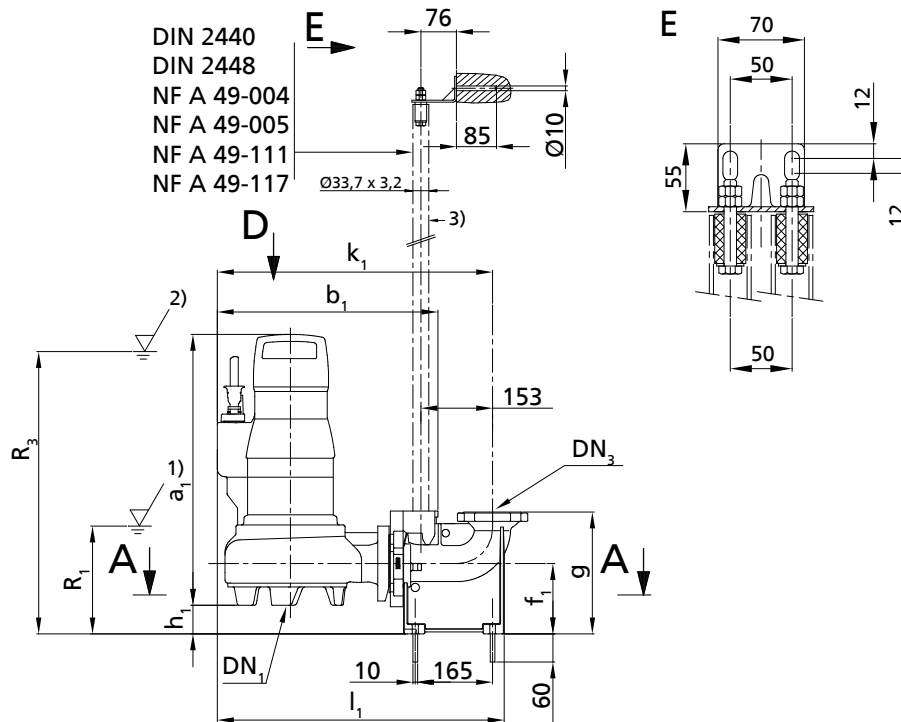
Clamped connection



Amarex N 65, stationary installation with twin guide rail arrangement

DN 3 = 65/65: DIN ISO ASME = standard - DN 3 = 65/80: DIN ISO = standard, ASME = variant

Twin guide rail arrangement

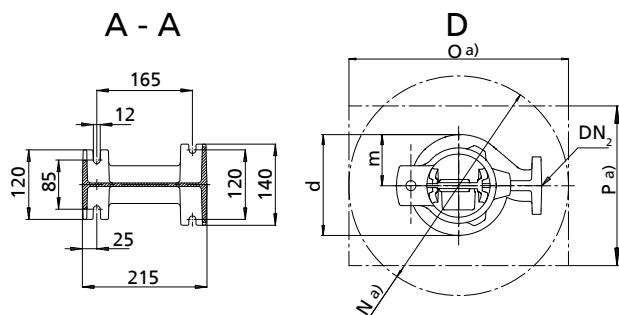


1)	Lowest switch-off point for automatic operation	2)	Minimum submergence for continuous operation
3)	Not included in KSB's scope of supply	a)	Minimum

Dimensions of pump and foundation [mm]

Size	Pump														Foundation		
	DN ₁	DN ₂	a ₁	b ₁	d	f ₁	g	h ₁	k ₁	l ₁	m	R ₁	R ₃	Z _r	N	O	P
65–170 F	65	65	578	468	251	150	260	61	588	613	127	234	639	4	550	550	400
65–220 F	65	65	518	454	265	150	260	63	574	599	142	241	581	4	550	550	400

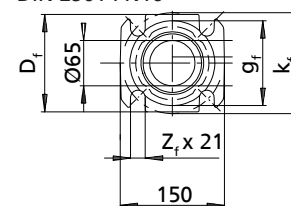
Installation in the sump



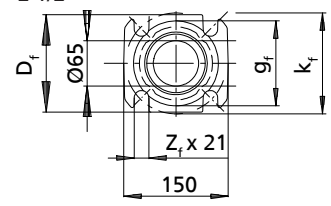
Flanged elbow DN

DN 65/65

ISO 7005 PN16
DIN 2501 PN16



ASME B16.1 class 125
2 1/2"



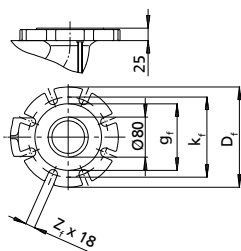
Dimensions of the flanged elbow DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	65	122	145	140	4
DIN 2501 PN16	80	133	160	200	8
ASME B16.1 class 125	65	122	140	140	4
	80	127	152,5	191	4

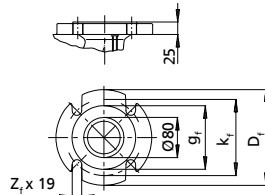
Extension elbow DN₃

DN 65/80

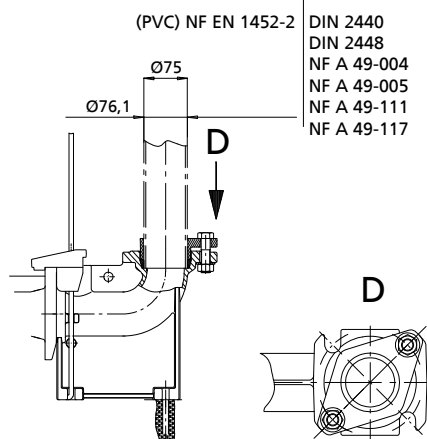
ISO 7005 PN16
DIN 2501 PN16



ASME B16.1 class 125
3"



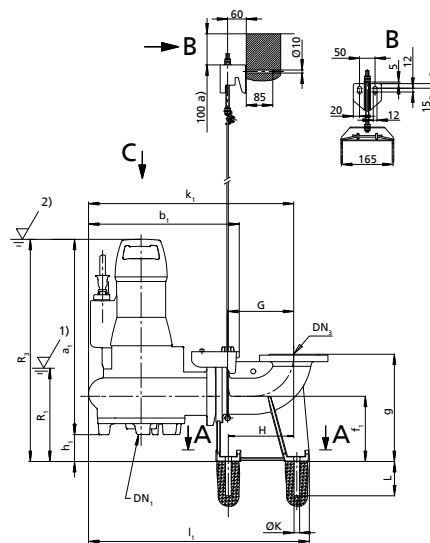
Clamped connection



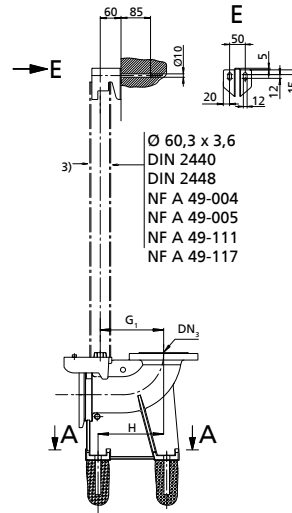
Amarex N 80 and 100, stationary installation with guide wire and single guide rail arrangement

DN 3 = 80/80: DIN ISO = standard, ASME = variant - DN 3 = 80/100 or 100/100: DIN ISO ASME = standard

Guide wire arrangement



Single guide rail arrangement



1)	Lowest switch-off point for automatic operation	2)	Minimum submergence for continuous operation
3)	Not included in KSB's scope of supply	a)	Minimum

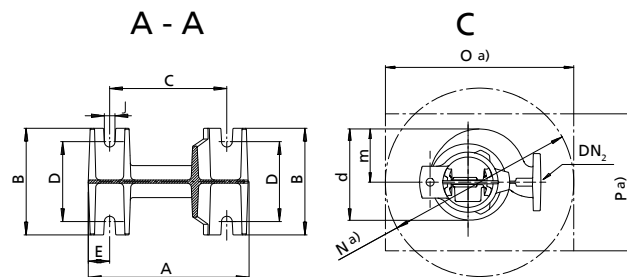
Pump dimensions [mm]

Size	Pump												
	DN ₁	DN ₂	a ₁	b ₁	d	f ₁	g	h ₁	k ₁	l ₁	m	R ₁	R ₃
80–220 F	80	80	582	478	322	200	320	103	604	694	176	262	685
80–220 D	-	80	602	478	322	200	320	86	604	694	176	262	688
100–220 F	100	100	603	476	318	210	345	98	641	691	169	280	701
100–220 D	-	100	628	476	318	210	345	76	641	691	169	280	704

Dimensions of the foundation in [mm]

Size	Foundation															
	DN ₃	A	B	C	D	E	G	G ₁	H	J	øK	L	N	O	P	
80–220 F/D	80	300	200	220	150	40	172,5	163	170	20	18	110	550	550	400	
80–220 F/D	100	300	200	220	150	40	172,5	163	170	20	18	110	550	550	400	
100–220 F/D	100	300	200	220	150	40	212,5	203	210	20	18	110	550	550	400	

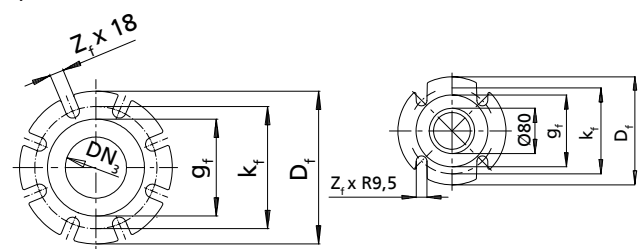
Installation in the sump



Flanged elbow DN

DN 80/80
ISO 7005 PN16
DIN 2501 PN16
DN 100/100
ISO 7005 PN16
DIN 2501 PN16
ASME B16.1 class 125
4"

DN 80/80
ASME B16.1 class 125
3"

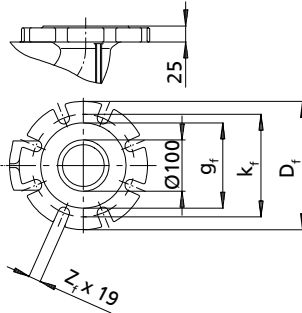


Extension elbow DN₃

DN 80/100

ISO 7005 PN16

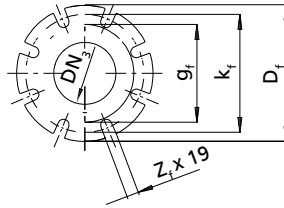
DIN 2501 PN16



DN 80/100

ASME B16.1 class 125

4"



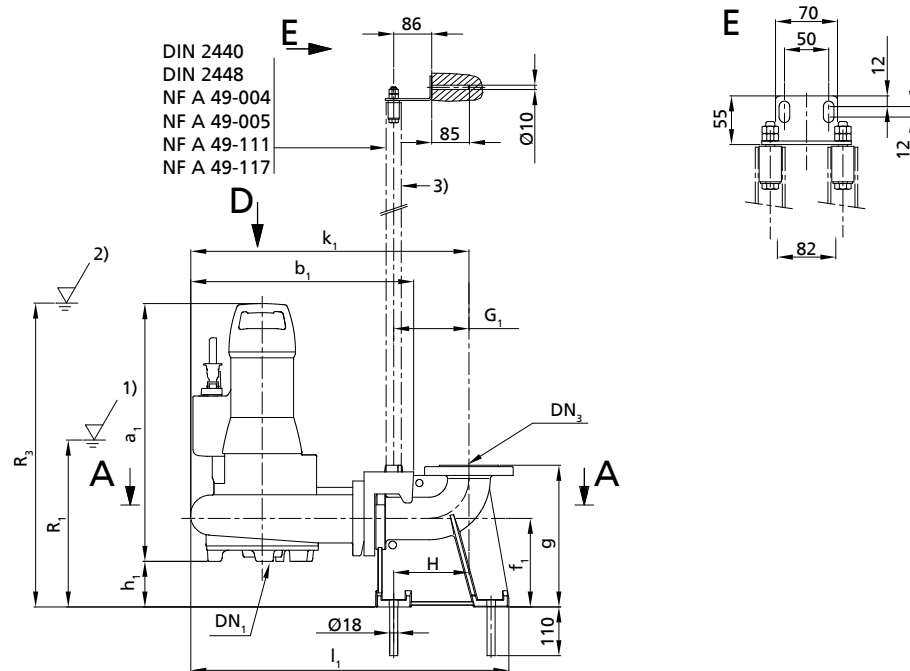
Dimensions of the flanged elbow DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	80	132	160	200	8
DIN 2501 PN16	100	156	180	220	8
ASME B16.1 class 125	80	127	152,5	191	4
	100	156	190,5	220	8

Amarex N 80 and 100, stationary installation with twin guide rail arrangement

DN 3 = 80/80: DIN ISO = standard, ASME = variant - DN 3 = 80/100 or 100/100: DIN ISO ASME = standard

Twin guide rail arrangement

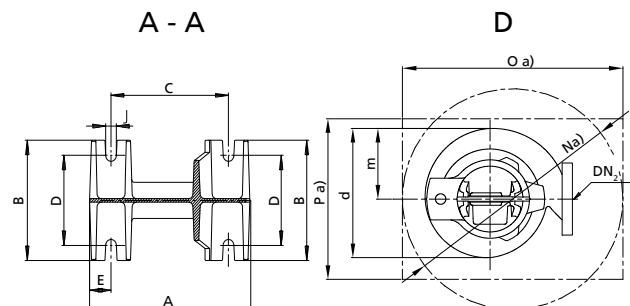


1)	Lowest switch-off point for automatic operation	2)	Minimum submergence for continuous operation
3)	Not included in KSB's scope of supply	a)	Minimum
b)	Maximum		

Pump dimensions [mm]

Size	Pump												
	DN ₁	DN ₂	a ₁	b ₁	d	f ₁	g	h ₁	k ₁	l ₁	m	R ₁	R ₃
80-220 F	80	80	582	506	322	200	320	103	630	720	176	262	685
80-220 D	-	80	602	506	322	200	320	86	630	720	176	262	688
100-220 F	100	100	603	529	318	210	345	98	674	724	169	280	701
100-220 D	-	100	628	529	318	210	345	76	674	724	169	280	704

Installation in the sump



Dimensions of the foundation in [mm]

Size	Foundation											
	DN ₃	A	B	C	D	E	G ₁	H	J	N	O	P
80-220 F	80	300	200	220	150	40	170	170	20	580	580	400
80-220 D	100	300	200	220	150	40	170	170	20	580	580	400
100-220 F	100	300	200	220	150	40	210	210	20	600	600	400
100-220 D	100	300	200	220	150	40	210	210	20	600	600	400

Flanged elbow DN

DN 80/80

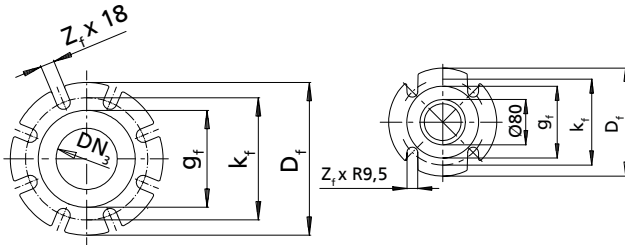
ISO 7005 PN16
DIN 2501 PN16

DN 100/100

ISO 7005 PN16
DIN 2501 PN16
ASME B16.1 class 125
4"

DN 80/80

ASME B16.1 class 125
3"



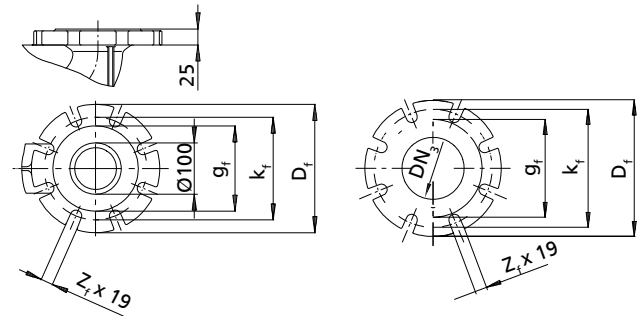
Extension elbow DN₃

DN 80/80

ISO 7005 PN16
DIN 2501 PN16

DN 80/100

ASME B16.1 class 125
4"



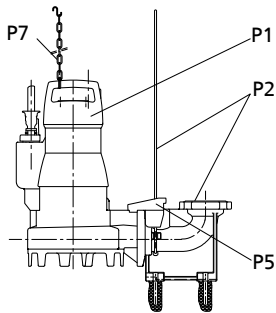
Dimensions of the flanged elbow DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	80	132	160	200	8
DIN 2501 PN16	100	156	180	220	8
ASME B16.1 150 lbs	80	127	152,5	191	4
	100	156	190,5	220	8

Installation types

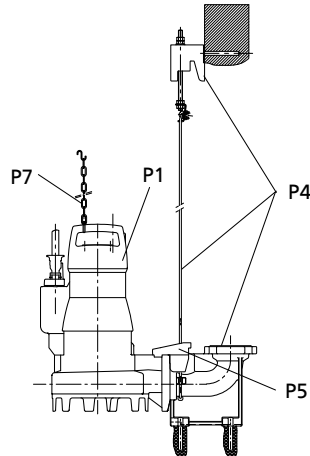
Overview of installation types

Installation type S - stationary wet installation



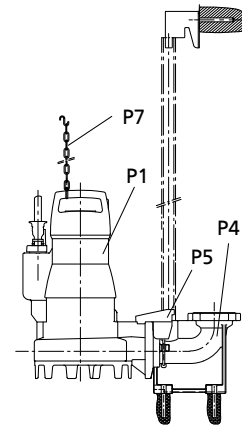
Guide hoop arrangement

P1: pump
P2: installation parts for guide hoop arrangement (Amarex N 50 and 65 only), installation depth = 1.5 m / 1.8 m / 2.1 m
P5: claw
P7: chain and shackle, length = 2 m



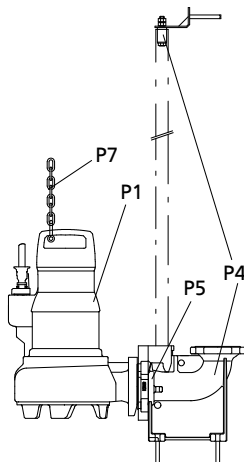
Guide wire arrangement

P1: pump
P4: installation parts for guide wire arrangement, installation depth = 4.5 m
P5: claw
P7: chain and shackle, length = 5 m



Single guide rail arrangement

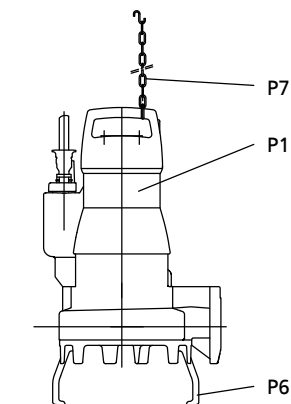
P1: pump
P4: installation parts for single guide rail arrangement
P5: claw
P7: chain and shackle, length = 5 m



Twin guide rail arrangement

P1: pump
P4: installation parts for twin guide rail arrangement
P5: claw and adapter
P7: chain and shackle, length = 5 m

Installation type P - transportable wet installation

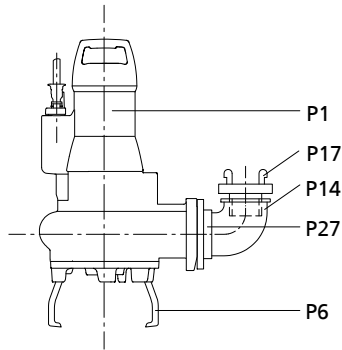


P1: pump
P6: foot
P7: chain and shackle, length = 5 m

Notes on installation

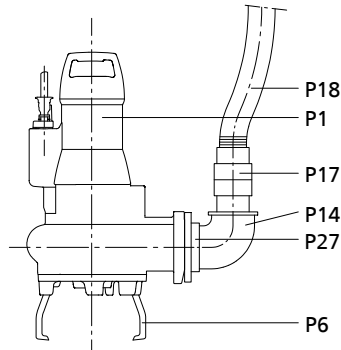
Suggested installation layouts for transportable pump sets

Size 50



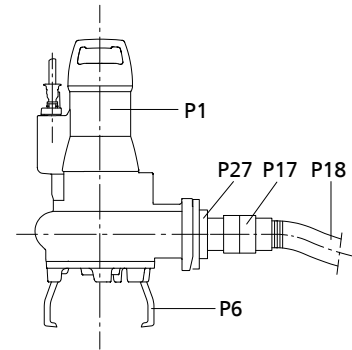
Suggestion 1

Vertical hose connection with elbow (P14) and Storz rigid coupling (P17) (quick connection)



Suggestion 2

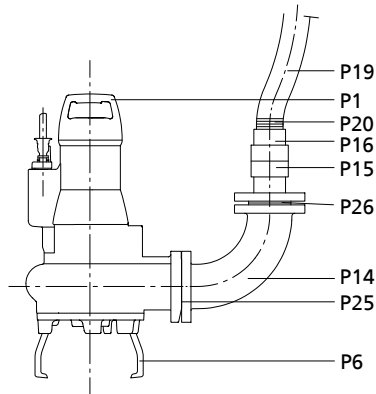
Vertical hose connection with plastic hose (P18)



Suggestion 3

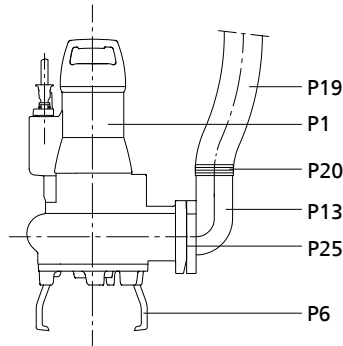
Horizontal hose connection (quick connection) with plastic hose (P18)

Sizes 65, 80, 100



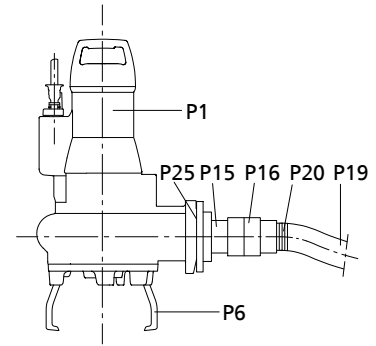
Suggestion 1

Vertical hose connection (quick connection) with plastic hose (P19) and hose clip (P20)



Suggestion 2

Vertical hose connection with plastic hose (P19), hose clip (P20) and connection elbow (P13)



Suggestion 3

Horizontal hose connection (quick connection) with plastic hose (P19), hose clip (P20), Storz hose coupling (P16), Storz rigid coupling (P15)

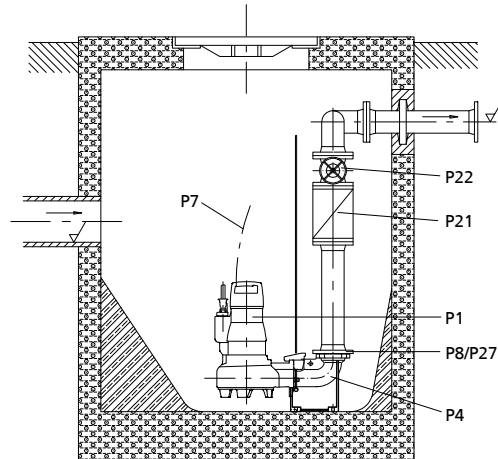
P1 to P27 (⇒ Page 42)

Suggested installation layouts for stationary pump sets

Sizes 50, 65, 80, 100

Guide hoop arrangement

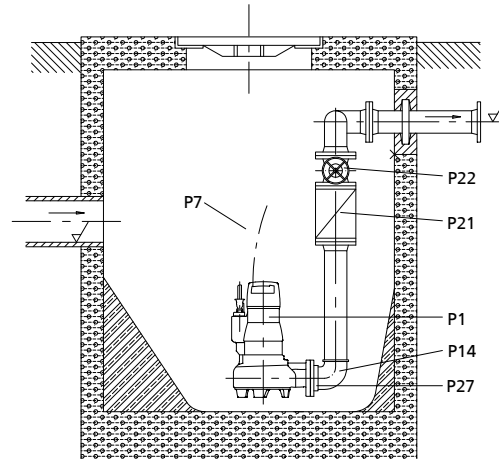
Amarex N S 50-172/F 50-170, S 50-222/F 50-220, F 65-170/65-220



Suggestion 1
Single-pump station
Flanged duckfoot bend

Suspended installation

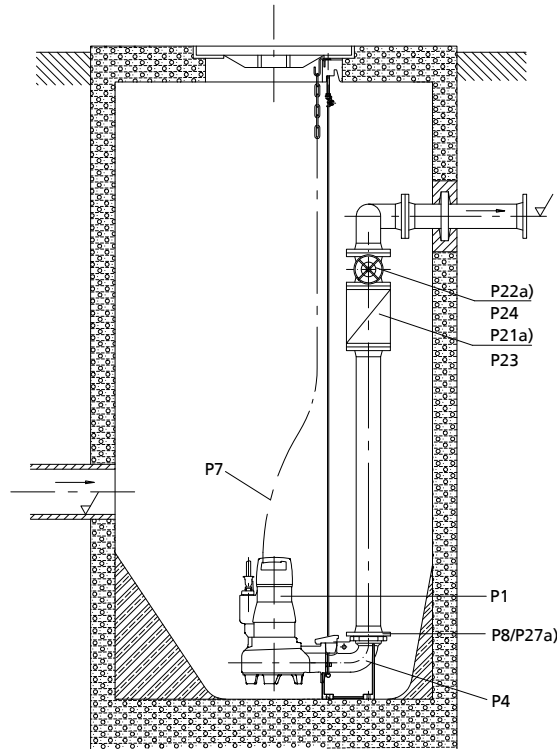
Amarex N S 50-172/F 50-170, S 50-222/F 50-220, F 65-170/65-220



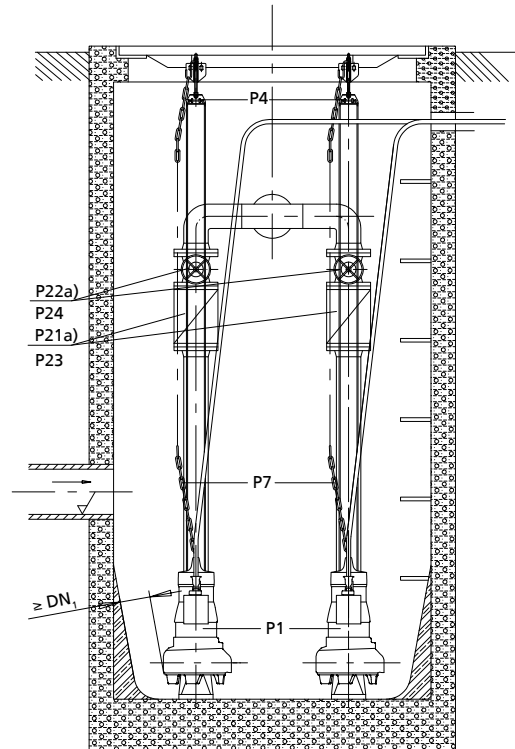
Suggestion 2
Single-pump station for 1.5 m installation depth
Direct connection to discharge pipe (suspended installation)

Guide wire, single guide rail or twin guide rail arrangement

Amarex N 50, 65, 80, 100



Suggestion 3
Either with guide wire, single guide rail or twin guide rail arrangement
Single-pump station for 4.5 m installation depth
Flanged duckfoot bend



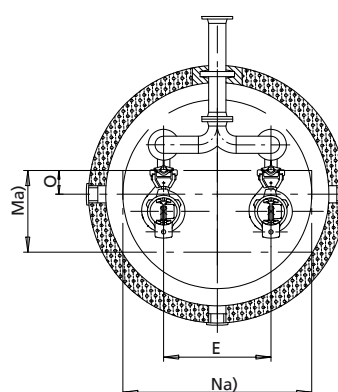
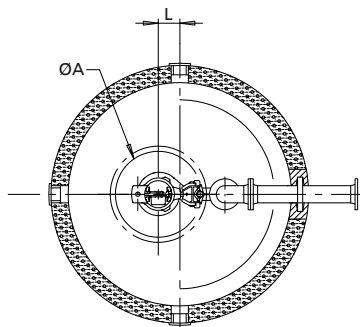
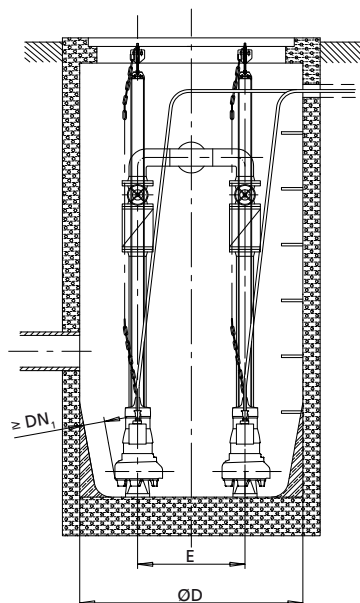
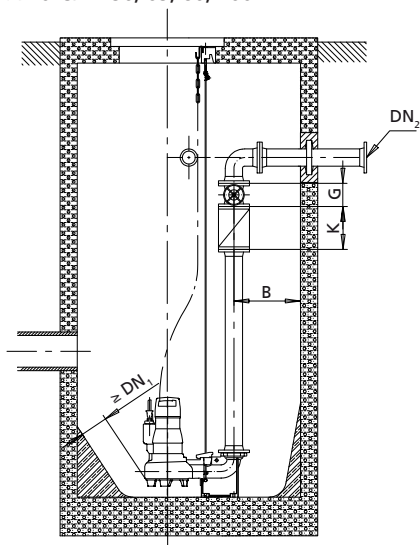
Suggestion 4
Either with guide wire, single guide rail or twin guide rail arrangement
Dual-pump station for 4.5 m installation depth
Flanged duckfoot bend

a) DN 50 only

Dimensions

Guide wire arrangement

Amarex N 50, 65, 80, 100



Single-pump station for 4.5 m installation depth
Flanged duckfoot bend

Dual-pump station for 4.5 m installation depth
Flanged duckfoot bend

a)	minimum
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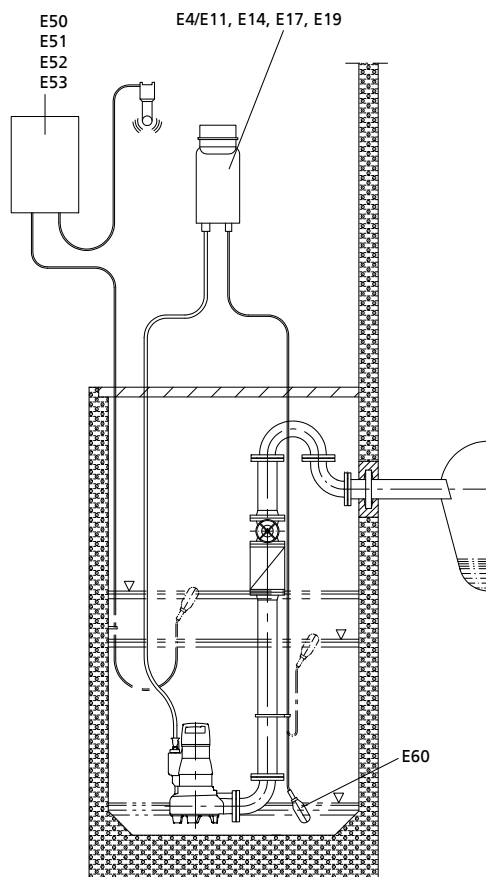
Dimensions [mm]

Amarex N		Ø A	B	Ø D	E	G	K	L	M	N	O	DN ₁	DN ₂
S 50-172 /	1 pump	625	165	1000	-	75	150	42	-	-	-	50	50
F 50-170	2 pumps	-	235	1000	300	75	150	-	550	700	200	50	50
S 50-222 /	1 pump	625	165	1000	-	75	150	42	-	-	-	50	50
F 50-220	2 pumps	-	235	1000	300	75	150	-	550	700	200	50	50
F 65-170 /	1 pump	625	175	1000	-	180	260	92	-	-	-	65	65
F 65-220	2 pumps	-	360	1200	600	180	260	-	550	1000	135	65	65
F 80-220 /	1 pump	625	200	1000	-	180	260	25	-	-	-	80	80
D 80-220	2 pumps	-	320	1200	600	180	260	-	600	1000	168	80	80
F 100-220	1 pump	625	200	1000	-	190	300	65	-	-	-	100	100
	2 pumps	-	320	1200	600	190	300	-	600	1000	128	100	100

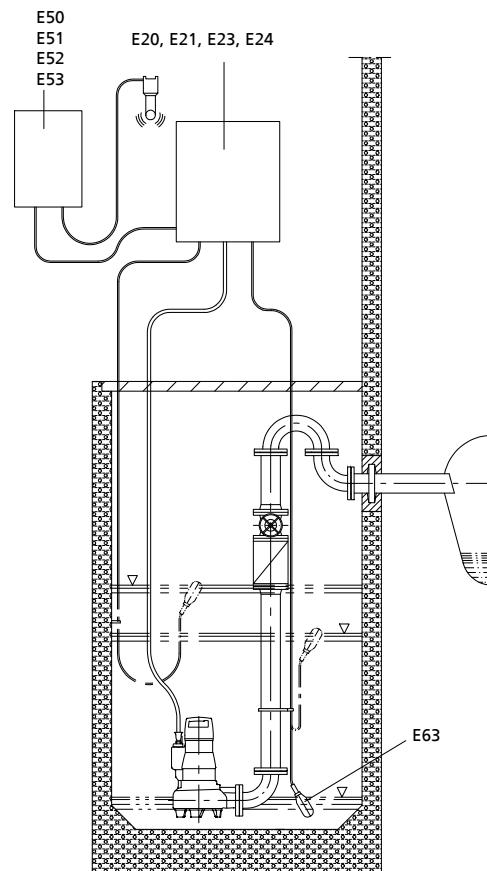
Suggested electrical installation layouts

Amarex N is available in explosion-proof and non-explosionproof design.

CEE motor protection switch (up to 4.0 kW)



Suggested installation layout No. 1
Non-explosionproof



Suggested installation layout No. 2
Explosion-proof

Scope of supply

Wet-installed stationary pump sets

- Pump set complete with power cables
- Claw with sealing element (O-ring) and mounting elements
- Lifting rope/lifting chain²⁰⁾
- Mounting bracket with mounting elements
- Duckfoot bend with mounting elements

- Guiding accessories
(guide rails are not included in KSB's scope of supply)

Wet-installed transportable pump sets

- Pump set complete with power cables
- Feet (and foot plate, if applicable)
- Lifting rope/lifting chain²¹⁾

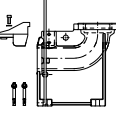
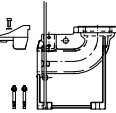
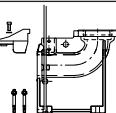
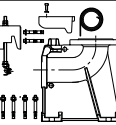
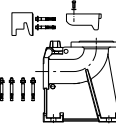
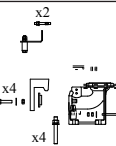
²⁰⁾ Optional

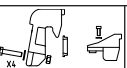
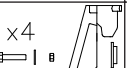
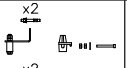
²¹⁾ Optional

Accessories

Installation parts for stationary pump sets

Overview of installation parts for stationary installation

	Item	Description	Sizes	Connection/ installation depth	Mat. No.	[kg]
Guide hoop arrangement						
	P2 + P5	Installation parts for stationary wet installation (guide hoop arrangement) Consisting of: DN 50 flanged duck foot bend, stainless steel anchor bolts, guide rail, claw with stainless steel screws/bolts, NOW WITHOUT CHAIN	Amarex DN 50, DN 3: DIN ISO ASME straight claw	Inst. depth 1.5 m	39022210	9
				Inst. depth 1.8 m	39022211	10
				Inst. depth 2.1 m	39022212	11
			Amarex N DN 50, DN 3: DIN ISO ASME inclined claw	Inst. depth 1.5 m	39022213	14
				Inst. depth 1.8 m	39022214	15
				Inst. depth 2.1 m	39022215	14
	P2 + P5	Installation parts for stationary wet installation (guide hoop arrangement) Consisting of: DN 65 flanged duck foot bend, stainless steel anchor bolts, guide rail, claw with stainless steel screws/bolts, NOW WITHOUT CHAIN	Amarex N DN 65, DN 3: DIN ISO ASME	Inst. depth 1.5 m	39020827	14.5
				Inst. depth 1.8 m	39020828	15.5
				Inst. depth 2.1 m	39020829	17
	P2 + P5	Installation parts for stationary wet installation (guide hoop arrangement) Consisting of: DN 65/80 flanged duck foot bend, stainless steel anchor bolts, guide rail, claw with stainless steel screws/bolts, NOW WITHOUT CHAIN	Amarex N DN 65/80, DN 3: DIN/ISO	Inst. depth 1.5 m	39020848	16
				Inst. depth 1.8 m	39020849	17
				Inst. depth 2.1 m	39020850	18.5
			Amarex N DN 65/80, DN 3: ASME	Inst. depth 1.5 m	39022255	16
				Inst. depth 1.8 m	39022256	17
				Inst. depth 2.1 m	39022257	16
Guide wire arrangement						
	P4 + P5	Installation parts for stationary wet installation, installation depth 4.5 m (guide wire installation) Consisting of: flanged duck foot bend, stainless steel anchor bolts, suspension bracket, mounting bracket, 10 m guide wire, claw with stainless steel screws/bolts, NOW WITHOUT CHAIN	Amarex N straight claw	DN 50, DN 3: DIN ISO ASME	39022196	13
			Amarex N inclined claw	DN 50, DN 3: DIN ISO ASME	39022200	18
			Amarex N	DN 65, DN 3: DIN ISO ASME	39020820	15
			Amarex N	DN 65/80, DN 3: DIN/ISO	39020834	19.1
			Amarex N	DN 65/80, DN 3: ASME	39020838	19.1
			Amarex N	DN 80, DN 3: DIN/ISO	39020988	35
			Amarex N	DN 80, DN 3: ASME	39020992	41
			Amarex N	DN 80/100, DIN ISO ASME	39021002	31.5
			Amarex N	DN 100, DIN ISO ASME	39021009	41
Single guide rail arrangement						
	P4 + P5	Installation parts for wet-installed stationary pump sets (single guide rail arrangement) for installation depth = 4.5 m Consisting of: flanged duckfoot bend, stainless steel anchor bolts, mounting bracket, claw with stainless steel screws/bolts, NOW WITHOUT CHAIN (guide rail not included in KSB's scope of supply)	Amarex N straight claw	DN 50, DN 3: DIN ISO ASME	39022204	12.5
			Amarex N inclined claw	DN 50, DN 3: DIN ISO ASME	39022207	18.5
			Amarex N	DN 65, DN 3: DIN ISO ASME	39021191	14.8
			Amarex N	DN 65/80, DN 3: DIN/ISO	39021194	18.7
			Amarex N	DN 65/80, DN 3: ASME	39021197	19.2
			Amarex N	DN 80, DN 3: DIN/ISO	39021200	26
			Amarex N	DN 80, DN 3: ASME	39021203	29.6
			Amarex N	DN 80/DN 100, DN 3: DIN ISO ASME	39021206	31
			Amarex N	DN 100, DN 3: DIN ISO ASME	39021209	26.8
Twin guide rail arrangement						
	P4 + P5	Installation parts for wet-installed stationary pump sets (twin guide rail arrangement) for installation depth = 4.5 m Consisting of: flanged duckfoot bend, stainless steel anchor bolts, mounting bracket, adapter, claw with stainless steel screws/bolts, NOW WITHOUT CHAIN (guide rails not included in KSB's scope of supply)	Amarex N inclined claw	DN 50, DN 3: DIN ISO ASME	39023002	14
			Amarex N straight claw	DN 65, DN 3: DIN ISO ASME	39023006	19.5
				DN 65/80, DN 3: DIN/ISO	39023009	22.8
				DN 65/80, DN 3: ASME	39023012	24.5
				DN 80, DN 3: DIN/ISO	39023018	32
				DN 80, DN 3: ASME	39023021	32
				DN 80/100, DN 3: DIN ISO ASME	39023024	34
				DN 100, DN 3: DIN ISO ASME	39023027	34

	Item	Description	Sizes	Connection/ installation depth	Mat. No.	[kg]	
Claw							
	P5	Claw Amarex N JL 1040 with stainless steel bolts	Amarex N straight claw	DN 50	39022248	1.1	
		Guide hoop arrangement for DN 50 and DN 65; guide wire and single guide rail arrangement for all DN	Amarex N inclined claw	DN 50	39022252	7	
			Amarex N	DN 65	39021018	1.9	
			Amarex N	DN 80/DN 100	39021020	3.1	
 	P5	Claw for Amarex N JL 1040 with stainless steel screws/bolts	Amarex N inclined claw	DN 50	39022990	6	
		Twin guide rails	Amarex N straight claw	DN 65	39022993	7.3	
			DN 80/DN 100	39022996	9.7		
			DN 100	39022999	14.7		
		Claw for Amarex JL1040 with stainless steel screws/bolts	Amarex straight claw	DN 50	39021016	1.1	
			Guide wire, single guide rail, guide hoop arrangement	Amarex inclined claw	DN 50	19551046	5
				Amarex	DN 65	39021018	1.9
				DN 80/DN 100	39021020	3.1	
Lifting bail							
 		Lifting bail made of stainless steel 1.4306 with A4-70 bolts, for lowering the pump set at an angle	Amarex N	DN 50	39022395	0.6	
				DN 65/DN 80/DN 100	39018004	1	
Conversion parts							
 		Parts for conversion to twin guide rail arrangement, consisting of: mounting bracket, stainless steel screws/bolts, adapter, anchor bolts Note: required for conversion of guide hoop, guide wire or single guide rail arrangement to twin guide rail arrangement Note: for twin guide rail arrangements a claw is mandatory.		DN 50/DN 65	39022984	1.4	
				DN 80/DN 100	39022987	2.5	

Off-standard designs on request.

Installation parts for transportable pump sets

Overview of installation parts for transportable pump sets

	Item	Description	Sizes	Mat. No.	[kg]
	P6	3 feet	Amarex N DN 50, 65, 80, 100	39022260	0.4
		Footplate incl. bolts (only for uneven mounting surfaces and in combination with feet)	Amarex N DN 50, 65, 80, 100	39022262	0.9

Chain for stationary and transportable pump sets

Overview of chains for stationary and transportable pump sets

Item	Description	Sizes		Mat. No.	[kg]
P7	Chain (1.4404) short-linked, tested and duly labelled to Directive 2006/42/EC (Machinery Directive), hook (1.4307), shackle (1.4401) Maximum load: 160 kg	2 m	Amarex N DN 50, 65	39023811	1.2
		3 m	Amarex N DN 50, 65, 80, 100	39023812	1.6
		5 m	Amarex N DN 50, 65, 80, 100	39023813	2.4
		10 m	Amarex N DN 50, 65, 80, 100	39023814	4.4

Item	Description	Sizes	Mat. No.	[kg]	
	Polypropylene lifting rope with shackle 1.4401 and hook 1.4571 ²²⁾	5 m	Amarex N DN 50, 65, 80, 100	39021975	2.5
	Shackle 1.4401, straight type, with stainless steel screw bolt	Amarex N	01019282	0.1	

Accessories for stationary and transportable pump sets

Overview of accessories for stationary and transportable pump sets

	Item	Description	Connection	Amarex N				Mat. No.	[kg]	
				50	65	80	100			
	P8	Flange for pipe coupling PN 10, at the flanged elbow, mating dimensions to PN 16	DN 50/R2" pipe	X	-	-	-	19551111	1.2	
			DN 65/R2½" pipe	-	X	-	-	39020184	1.2	
	P9	PVC adapter for hose connection, with 1 hose clip, plastic hose of inner diameter 63 item 19	R2"	X	-	-	-	11191498	0.3	
	P13	Connection elbow with flange/hose connection made of JL 1040, grey cast iron PN 16, DIN 2501, including joint ring and hose clip; for DN 100 also with fixing bolts To be used for flange connections item 25 / item 26 (not for DN 100).	DN 65/B 75	-	X	-	-	19135655	6	
			DN 80/B 75	-	-	X	-	19131746	6.6	
			DN 100/A 110	-	-	-	X	19139718	10	
	P14	Elbow with external/internal thread (to be used for flange connections item 27 /item 30), galvanised grey cast iron	R2"	X	-	-	-	00241966	0.3	
		Connection elbow with flanges PN 16, DIN 2501 (to be used for flange connections item 25 / item 26), grey cast iron	DN 65/65	-	X	-	-	00265480	11	
			DN 65/80	-	X	-	-	25198402	8	
			DN 80/80	-	-	X	-	11150856	10	
			DN 100/100	-	-	-	X	25145802	14.4	
	P15	Storz rigid coupling with flange, to DIN 2501, drilled to PN 16 (to be used for flange connections item 25 / item 26), aluminium/steel	DN 65/B 75	-	X	-	-	18040148	3.5	
			DN 80/B 75	-	-	X	-	18072642	3.5	
			DN 100/A 110	-	-	-	X	18060162	5	
	P16	Storz hose coupling, aluminium alloy 2 hose clips, item 20, are required for hose mounting (for plastic hoses B 75 and A 110, item 19)	C 52 (DIN 14321)	X	-	-	-	00524551	0.3	
			B 75 (DIN 14322)	-	X	X	-	00520454	0.7	
			A 110 (DIN 14323)	-	-	-	X	00522313	1.5	
	P17	Storz rigid coupling with external thread, DIN ISO 228/1	C 52/G 2 A	X	-	-	-	00524370	0.2	
			B 75 - G 2½	-	X	-	-	00524371	0.4	
	P18	Plastic hose DN 50, DIN 14811, with integrated C couplings	C 52-5 m	X	-	-	-	00522262	2.3	
			C 52-10 m	X	-	-	-	00522263	4.2	
			C 52-20 m	X	-	-	-	00522264	5.7	
	P19	Plastic hose Without coupling, DIN 14 811	Inside diameter 63	5 m	X	-	-	-	39018688	1.7
				10 m	X	-	-	-	39018689	3.4
				20 m	X	-	-	-	39018690	6.8
			B 75	5 m	-	X	X	-	39019064	2
				20 m	-	X	X	-	39019066	8
				30 m	-	X	X	-	39019071	12
			Inside diameter 80	5 m	-	-	X	-	39018691	2.2
				10 m	-	-	X	-	39019062	4.3
			A 110	5 m	-	-	-	X	39019067	4.7
				10 m	-	-	-	X	39019068	9.3
30 m	-	-		-	X	39019070	27.9			

²²⁾ Increase quantity for larger installation depths.

	Item	Description	Connection	Amarex N				Mat. No.	[kg]
				50	65	80	100		
	P20	Hose clip DIN 3017, chrome steel	B 50 ²³⁾	X	-	-	-	39000515	0.025
			B 75	-	X	X	-	00109515	0.04
			AL 110 - 120 B ²⁴⁾	-	-	-	X	00520853	0.1
	P21	RK swing check valve Plastic, EN 12 050-4, with internal thread ISO 7/1, full port and drain plug; cannot be used for pumped drainage	Rp 2	X	-	-	-	01009773	0.5
	P22	Socket gate valve CuZn PN 10-12 DIN 3352	Rp 2	X	-	-	-	00411503	1.1
			Rp 2½	-	X	-	-	39000507	1.7
	P23	KSB swing check valve, grey cast iron With full port, backwash device, flanges drilled to DIN 2501, PN 16 (not for lifting units)	DN 65	-	X	-	-	48829253	16.2
			DN 80	-	-	X	-	48829254	21.5
			DN 100	-	-	-	X	48829255	29
	P24	ECOLINE GTR-16P gate valve, grey cast iron, PN 16, flanges drilled to ISO 7005/DIN 2501	DN 65	-	X	-	-	49709579	15
			DN 80	-	-	X	-	49709580	22
			DN 100	-	-	-	X	49709581	26.5
	P25	Set of installation accessories for a flange connection, discharge nozzle; item 13, 14 or 15 Consisting of: 4 hexagon head bolts with nuts and 1 sealing element		X	-	-	-	39021944	0.8
				-	X	-	-	19551115	0.8
				-	-	X	-	19551100	0.8
				-	-	-	X	19551113	0.8
	P26	Set of installation accessories for a flange connection, Consisting of: 8 hexagon head bolts with nuts and 1 sealing element		-	-	X	-	19551114	0.8
				-	-	-	X	19551116	0.8
	P27	Threaded flange for guide hoop arrangement (P2) and guide wire arrangement (P4) Consisting of: flange, 4 hexagon head bolts with nuts and discs, and 1 sealing element	DN 50/Rp 2	X	-	-	-	19551353	2
			DN 65/Rp 2½	-	X	-	-	39021943	2.9
	P28	Hand pump, wall mounting, grey cast iron, suction-side connection Rp 1½		X	X	X	X	00520485	12
	P30	Adapter with reduced external thread M4 EN 1042K	2 x 1¼	-	-	-	-	01135663	0.4

²³⁾ For plastic hose Ø 63 item 19

²⁴⁾ 2 units required

Electrical accessories

Control units for Amarex N without ATEX

i Valid for all countries except France.
Control units for Amarex N without ATEX

Item	Description	Power min [A]	Power max [A]	Type	Mat. No.	[kg]
E 4	Multi-functional plug, type Hyper, with motor protection relay CEE plug	2,6	3,7	Hyper 37.1	19071492	1
		3,7	5,5	Hyper 55.1	19071493	1
		5,5	8,0	Hyper 80.1	19071494	1
		8,0	11,5	Hyper 115.1	19071495	1
LevelControl Basic 2 control unit for single-pump station						
E 11	For float switch or 4...20 mA sensor, optional with master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 ^{DFNO} 040	19073763	4.5
		4,0	6,3	BC1 400 ^{DFNO} 063	19073764	4.5
		6,3	10,0	BC1 400 ^{DFNO} 100	19073765	4.5
E 14	Pneumatic, optional master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 ^{DPNO} 040	19073768	4.5
		4,0	6,3	BC1 400 ^{DPNO} 063	19073769	4.5
		6,3	10,0	BC1 400 ^{DPNO} 100	19073770	4.5
E 17	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 ^{DLNO} 040	19073818	12
		4,0	6,3	BS1 400 ^{DLNO} 063	19073819	12
		6,3	10,0	BS1 400 ^{DLNO} 100	19073820	12
E 19	Bubbler control in BC variant, 400 x 278 x 120 mm Only to be used for connections with neutral conductor! Installation option O1 (master switch) not possible! 400 x 278 x 120 mm	2,5	4,0	BC1 400 ^{DLNO} 040	19075148	4.5
		4,0	6,3	BC1 400 ^{DLNO} 063	19075149	4.5
		6,3	10,0	BC1 400 ^{DLNO} 100	19075150	4.5
LevelControl Basic 2 control unit for dual-pump stations						
E 31	For float switch or 4...20 mA sensor, optional master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 ^{DFNO} 040	19073777	4.7
		4,0	6,3	BC2 400 ^{DFNO} 063	19073778	4.7
		6,3	10,0	BC2 400 ^{DFNO} 100	19073779	4.7
E 34	Pneumatic, optional master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 ^{DPNO} 040	19073782	4.7
		4,0	6,3	BC2 400 ^{DPNO} 063	19073783	4.7
		6,3	10,0	BC2 400 ^{DPNO} 100	19073784	4.7
E 37	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 ^{DLNO} 040	19073860	13
		4,0	6,3	BS2 400 ^{DLNO} 063	19073861	13
		6,3	10,0	BS2 400 ^{DLNO} 100	19073862	13
E 39	Bubbler control for BC version Only to be used for connections with neutral conductor! Installation option O1 (master switch) not possible! 400 x 278 x 120 mm	2,5	4,0	BC2 400 ^{DLNO} 040	19075151	4.7
		4,0	6,3	BC2 400 ^{DLNO} 063	19075152	4.7
		6,3	10,0	BC2 400 ^{DLNO} 100	19075153	4.7

i Only valid for France!
Control units for Amarex N without ATEX

Item	Description	Power min [A]	Power max [A]	Type	Mat. No.	[kg]
E 4	Multi-functional plug, type Hyper, with motor protection relay CEE plug	2,6	3,7	Hyper 37.1	19071492	1
		3,7	5,5	Hyper 55.1	19071493	1
		5,5	8,0	Hyper 80.1	19071494	1
		8,0	11,5	Hyper 115.1	19071495	1
	LevelControl Basic 2 control unit for single-pump stations					
E11	For float switch or 4...20 mA sensor, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 ^{DFNO} 040 02	19073878	4.5
		4,0	6,0	BC1 400 ^{DFNO} 063 02	19073879	4.5
		6,0	10,0	BC1 400 ^{DFNO} 100 02	19073880	4.5
E14	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 ^{DPNO} 040 02	25)	25)
		4,0	6,3	BC1 400 ^{DPNO} 063 02	25)	25)
		6,3	10,0	BC1 400 ^{DPNO} 100 02	25)	25)
E17	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 ^{DLNO} 040 02	25)	25)

25) See KSB Easy Select

Item	Description	Power min [A]	Power max [A]	Type	Mat. No.	[kg]
		4,0	6,3	BS1 400 DLNO 063 02	25)	25)
		6,3	10,0	BS1 400 DLNO 100 02	25)	25)
	LevelControl Basic 2 control unit for dual-pump stations					
E11	For float switch or 4...20 mA sensor, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 DFNO 040 02	19073888	4.7
		4,0	6,3	BC2 400 DFNO 063 02	19073889	4.7
		6,0	10,0	BC2 400 DFNO 100 02	19073890	4.7
E14	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 DPNO 040 02	25)	25)
		4,0	6,3	BC2 400 DPNO 063 02	25)	25)
		6,3	10,0	BC2 400 DPNO 100 02	25)	25)
E17	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 DLNO 040 02	25)	25)
		4,0	6,3	BS2 400 DLNO 063 02	25)	25)
		6,3	10,0	BS2 400 DLNO 100 02	25)	25)

Alarm switchgears for Amarex N without ATEX

Overview of alarm switchgears for Amarex N without ATEX

Item	Description	Mat. No.	[kg]
E 50	AS 0Buzzer/switching input	29128401	0.5
E 51	AS 2Buzzer/switching input, mains-dependent with volt-free signalling contact	29128422	0.5
E 52	AS 4Buzzer/switching input, mains-independent (battery-backed) with volt-free signalling contact and self-charging power supply unit (for 5 hours' operation in the event of power failure)	29128442	0.5
E 53	AS 5Alarm equipment (e.g. horn), volt-free contact/battery buffer	00530561	1.7
E 55	AS 1Buzzer/moisture sensor	00533740	0.9

Control unit accessories for Amarex N without ATEX

Overview of control unit accessories for Amarex N without ATEX

	Item	Description		Mat. No.	[kg]
	E 60	Float switch with free cable end (NO contact) Switch housing made of polypropylene (max. fluid temperature 70 °C), circuit closed in upper float position, connection cable (H 07 RN-F3G1)	3 m	11037742	0.5
			5 m	11037743	0.8
			10 m	11037744	1.3
			15 m	11037745	1.8
			20 m	11037746	2.4
			25 m	11037747	2.9
			30 m	11037748	3.4
	E 62	Float switch with free cable end (NC contact), circuit open in upper float position, connection cable (H 07 RN-F3G1)	5 m	11037756	0.8
			10 m	11037757	1.4
			20 m	11037758	2.6
	E 64	Leakage sensor F 1	3 m	19072366	0.2
	E65	Open pressure bell set - pneumatic and bubbler control with polyamide tube 8 x 1 mm	Tube length 10 m	19071721	1.2
			Tube length 20 m	19071837	2
			Tube length 50 m	19074200	2.5
	E66	Closed pressure bell set - pneumatic with polyamide tube 8 x 3 mm	Tube length 10 m	19071722	3.5
			Tube length > 10 m		
	E70	Horn, 12 V DC, 105 dB(A), 1.2 W		01086547	0.1
	E71	Alarm combination, 12 V DC		01139930	0.1
	E72	Alarm strobe light, 12 V DC		01056355	0.3
	E73	PC service tool		47121210	0.2
	E90	Rechargeable battery retrofit kit for LevelControl Basic 2, for powering the electronics, the float switches or internal pressure sensor and the alarm equipment (buzzer, horn, alarm combination), for single-pump and dual-pump stations	For type BC, consisting of 2 rechargeable batteries 6 V, 1.3 Ah and charging unit	19074194	0.8
	E91		For type BS, consisting of 1 rechargeable battery 12 V, 1.2 Ah and charging unit	19074199	1

Control units for Amarex N with ATEX

i The control units are not explosion-proof, which means that they must not be operated in potentially explosive atmospheres.

i For ATEX-compliant variants, intrinsic safety barriers and an ATEX level switch (float switch) must be fitted. Selection via KSB EasySelect.

i **Valid for all countries except France.**

Control unit accessories for Amarex N with ATEX

Item	Description	Power min [A]	Power max [A]	Type	Mat. No.	[kg]
LevelControl Basic 2 control unit for single-pump stations						
E 20	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS1 400 ^{DFEO} 040	19073800	12
		4,0	6,3	BS1 400 ^{DFEO} 063	19073801	12
		6,3	10,0	BS1 400 ^{DFEO} 100	19073802	12
E 21	Pneumatic, optional master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 ^{DPEO} 040	19073771	4.5
		4,0	6,3	BC1 400 ^{DPEO} 063	19073772	4.5
		6,3	10,0	BC1 400 ^{DPEO} 100	19073773	4.5
E 23	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 ^{DLEO} 040	19073821	12
		4,0	6,3	BS1 400 ^{DLEO} 063	19073822	12
		6,3	10,0	BS1 400 ^{DLEO} 100	19073823	12
E 24	Bubbler control for BC version	2,5	4,0	BC1 400 ^{DLEO} 040	19075154	4.5
	Only to be used for connections with neutral conductor! Installation option O1 (master switch) not possible! 400 x 278 x 120 mm	4,0	6,3	BC1 400 ^{DLEO} 063	19075155	4.5
		6,3	10,0	BC1 400 ^{DLEO} 100	19075156	4.5
LevelControl Basic 2 control unit for dual-pump stations						
E 40	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS2 400 ^{DFEO} 040	19073842	13
		4,0	6,3	BS2 400 ^{DFEO} 063	19073843	13
		6,3	10,0	BS2 400 ^{DFEO} 100	19073844	13
E 41	Pneumatic, optional master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 ^{DPEO} 040	19073785	4.7
		4,0	6,3	BC2 400 ^{DPEO} 063	19073786	4.7
		6,3	10,0	BC2 400 ^{DPEO} 100	19073787	4.7
E 43	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 ^{DLEO} 040	19073863	13
		4,0	6,3	BS2 400 ^{DLEO} 063	19073864	13
		6,3	10,0	BS2 400 ^{DLEO} 100	19073865	13
E 44	Bubbler control for BC version	2,5	4,0	BC2 400 ^{DLEO} 040	19075157	4.7
	Only to be used for connections with neutral conductor! Installation option O1 (master switch) not possible! 400 x 278 x 120 mm	4,0	6,3	BC2 400 ^{DLEO} 063	19075158	4.7
		6,3	10,0	BC2 400 ^{DLEO} 100	19075159	4.7

i **Only valid for France!**

Control unit accessories for Amarex N with ATEX

Item	Description	Power min [A]	Power max [A]	Type	Mat. No.	[kg]
	LevelControl Basic 2 control unit for single-pump stations					
E20	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS1 400 ^{DFEO} 040 02	26)	26)
		4,0	6,3	BS1 400 ^{DFEO} 063 02	26)	26)
		6,3	10,0	BS1 400 ^{DFEO} 100 02	26)	26)
E21	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 ^{DPEO} 040 02	26)	26)
		4,0	6,3	BC1 400 ^{DPEO} 063 02	26)	26)
		6,3	10,0	BC1 400 ^{DPEO} 100 02	26)	26)
E23	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 ^{DLEO} 040 02	26)	26)
		4,0	6,3	BS1 400 ^{DLEO} 063 02	26)	26)
		6,3	10,0	BS1 400 ^{DLEO} 100 02	26)	26)
	LevelControl Basic 2 control unit for dual-pump stations					
E40	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS2 400 ^{DFEO} 040 02	26)	26)
		4,0	6,3	BS2 400 ^{DFEO} 063 02	26)	26)

26) See KSB Easy Select

Item	Description	Power min [A]	Power max [A]	Type	Mat. No.	[kg]
		6,3	10,0	BS2 400 ^{DPEO} 100 02	26)	26)
41	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 ^{DPEO} 040 02	26)	26)
		4,0	6,3	BC2 400 ^{DPEO} 063 02	26)	26)
		6,3	10,0	BC2 400 ^{DPEO} 100 02	26)	26)
E43	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 ^{DLEO} 040 02	26)	26)
		4,0	6,3	BS2 400 ^{DLEO} 063 02	26)	26)
		6,3	10,0	BS2 400 ^{DLEO} 100 02	26)	26)

Control unit accessories for Amarex N with ATEX

Overview of accessories for Amarex N with ATEX

	Item	Description		Mat. No.	[kg]
	E 63	Float switch with free cable end (NO contact) With declaration of compliance with explosion protection standards Power cable (H 07 RN-F3G1)	5 m	01148226	0.7
			10 m	01148247	1
			20 m	01148248	2
	E65	Open pressure bell set - pneumatic and bubbler control with polyamide tube 8 x 1 mm	Tube length 10 m	19071721	1.2
			Tube length 20 m	19071837	2
			Tube length 50 m	19074200	2.5
	E66	Closed pressure bell set - pneumatic with polyamide tube 8 x 3 mm	Tube length 10 m	19071722	3.5
			Tube length > 10 m		
	E70	Horn, 12 V DC, 105 dB(A), 1.2 W, not explosion-proof		01086547	0.1
	E71	Alarm combination, 12 V DC, not explosion-proof		01139930	0.1
	E72	Alarm strobe light, 12 V DC, not explosion-proof		01056355	0.3
	E73	PC service tool		47121210	0.2
	E90	Rechargeable battery retrofit kit for LevelControl Basic 2, for powering the electronics, the float switches or internal pressure sensor and the alarm equipment (buzzer, horn, alarm combination), for single-pump and dual-pump stations	For type BC, consisting of 2 rechargeable batteries 6 V, 1.3 Ah and charging unit	19074194	0.8
	E91		For type BS, consisting of 1 rechargeable battery 12 V, 1.2 Ah and charging unit	19074199	1



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